Formal versus Informal Finance: Evidence from China

Meghana Ayyagari
School of Business, George Washington University

Asli Demirgüç-Kunt
World Bank

Vojislav Maksimovic
Robert H. Smith School of Business at the University of Maryland

The fast growth of Chinese private sector firms is taken as evidence that informal finance can facilitate firm growth better than formal banks in developing countries. We examine firm financing patterns and growth using a database of twenty-four hundred Chinese firms. While a relatively small percentage of firms utilize bank loans, bank financing is associated with faster growth whereas informal financing is not. Controlling for selection, we find that firms with bank financing grow faster than similar firms without bank financing and that our results are not driven by bank corruption or the selection of firms that have accessed the formal financial system. Our findings question whether reputation and relationship-based financing are responsible for the performance of the fastest-growing firms in developing countries. (*JEL* G21, G30, O16, O17)

Financial development has been shown to be associated with faster growth and improved allocative efficiency.[1] While the research focus has been on formal financial institutions, the literature has recognized the existence and role played by informal financial systems, especially in developing economies.[2]

We would like to thank Franklin Allen, Amit Bubna, Joseph Fan, Patrick Honohan, Jianjun Li, Jordan Siegel, Colin Xu, Jiawen Yang, Bernard Yeung, and seminar participants at AFE New Orleans 2008, Indian School of Business, University of Amsterdam, McGill University, the Chinese University of Hong Kong's (CUHK) Conference on Contemporary Issues of Firms and Institutions, World Bank Small Business Finance Conference, and Washington University for their suggestions and comments. This article's findings, interpretations, and conclusions are entirely those of the authors and do not necessarily represent the views of the World Bank, its Executive Directors, or the countries they represent. This research was supported by the National Science Foundation, NSF Grant #SES-0550573, 0550454. Send correspondence to Meghana Ayyagari, 2201 G Street NW, Funger Hall 401, Washington, DC 20052; telephone: (202) 994-1292. E-mail: ayyagari@gwu.edu.

[1] This literature includes cross-country studies (King and Levine 1993; Levine and Zervos 1998; Levine, Loayza, and Beck 2000; Bekaert, Harvey, and Lundblad 2001, 2005), firm level-studies (Demirgüç-Kunt and Maksimovic 1998; Love 2003; Beck, Demirgüç-Kunt, and Maksimovic 2005), industry-level studies (Rajan and Zingales 1998; Wurgler 2000; Beck et al. 2008), and individual country case studies (Guiso, Sapienza, and Zingales 2004a in Italy and Bertrand, Schoar, and Thesmar 2007 in France).

[2] By informal financial institutions, we refer to loans from moneylenders, landlords, and family who base the financial transaction on business/personal relationships, as well as loans from institutions such as credit cooperatives and savings and credit associations in certain countries that provide financial intermediation between savers and borrowers but do not rely on the state to enforce contractual legal obligations. See Section 1 for details.

© The Author 2010. Published by Oxford University Press on behalf of The Society for Financial Studies. All rights reserved. For Permissions, please e-mail: journals.permissions@oxfordjournals.org.
doi:10.1093/rfs/hhq030 Advance Access publication May 6, 2010

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

The dominant view is that informal financial institutions play a complementary role to the formal financial system by servicing the lower end of the market. Informal financing typically consists of small, unsecured, short-term loans restricted to rural areas, agricultural contracts, households, individuals, or small entrepreneurial ventures. Informal financial institutions rely on relationships and reputation and can more efficiently monitor and enforce repayment from a class of firms than commercial banks and similar formal financial institutions can.[3] According to this view, however, informal financial systems cannot substitute for formal financial systems, because their monitoring and enforcement mechanisms are ill equipped to scale up and meet the needs of the higher end of the market.[4]

In developed markets such as the United States, there is a direct parallel with the prevalence of angel financing where high net-worth individuals, "angel investors," provide initial funding to young new firms with modest capital needs until they are able to receive more formal venture capital financing (Fenn, Liang, and Prowse 1997).[5] Guiso, Sapienza, and Zingales (2004a) show that social capital impacts regional financial development in Italy and is particularly important when legal enforcement is weak. Gomes (2000) shows that minority shareholders invest in initial public offerings even in environments with poor investor protection rights, since controlling shareholders commit implicitly not to expropriate them because of reputation concerns. A large literature in finance has focused on the role of relationships in affecting access to credit. Petersen and Rajan (1994) and Berger and Udell (1995) focus on the information in firm–creditor relationships in controlling access to bank loans, while Hellman, Lindsey, and Puri (2008) explore the impact of private information in bank venture capital relationships on bank lending decisions. Garmaise and Moskowitz (2003) show that banks and brokers in the commercial real estate market develop informal networks that have a significant effect on availability of finance to the brokers' clients.

While the above work shows the existence of informal networks alongside formal systems, an important extension of this literature has been recent work by Tsai (2002), Allen, Qian, and Qian (2005), and Linton (2006). These authors argue that private sector firms in China, despite facing weaker legal protections and poorer access to finance than firms in the state and listed sectors, are the fastest growing due to their reliance on alternative financing and

[3] A large economics literature has also argued that informal institutions have a comparative advantage in enforcement capacity and monitoring (the peer monitoring view as in Stigltiz 1990 and Arnott and Stiglitz 1991). Theoretically, the informal sector has been modeled as both a competitor to its formal counterpart (Bell, Srinivasan, and Udry 1997; Jain 1999; Varghese 2005), as well as a channel of formal funds where informal lenders acquire formal funds to service entrepreneurs' financing needs (Floro and Ray 1997; Bose 1998; Hoff and Stiglitz 1998).

[4] Ayyagari, Demirguc-Kunt, and Maksimovic (2009) argue that informal systems cannot substitute for formal monitoring at the higher end of the market and show that informal financing is associated with greater tax evasion than formal financing.

[5] While angel financing is important for early stage start-ups, Lerner (1998) argues that there is no evidence that angel financing generates positive financial returns or improves social welfare.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

governance mechanisms.[6] Allen, Qian, and Qian (2005) further suggest that China may be an important counterexample to the law and finance literature's focus on formal systems, since the fastest-growing Chinese firms rely on alternative financing channels rather than formal external finance.

In this article, we use detailed firm-level survey data on twenty-four hundred firms in China to investigate which of the two views are consistent with the operation of the informal sector in China. Is the informal sector associated with high growth and profit reinvestment and does it serve as a substitute to the formal financial system or does the informal sector primarily serve the lower end of the market? To answer these questions, we first investigate whether Chinese firms' financing patterns are different when compared to those of similar firms in other countries. Next, we explore how their financing patterns, both formal and informal, vary across different types of firms, across regions. We then study how bank finance and financing from informal sources are associated with firm sales growth, productivity growth, and profit reinvestment.

We address these issues using a new data source, the Investment Climate Survey,[7] a major firm-level survey conducted in China in 2003 and led by the World Bank. The survey has information on financing choices for firms across eighteen different cities. One of the strengths of the survey is its coverage of small and medium enterprises (SMEs). Hence, in addition to information on commercial financing sources such as bank finance, the survey also includes information on sources of financing associated with small firm finance such as trade credit and finance from informal sources such as a moneylender or an informal bank or other financing sources.

We find that 20% of firm financing in our sample is sourced from banks, which is comparable to the use of bank financing in other developing countries, such as India, Indonesia, Brazil, Bangladesh, Nigeria, and Russia. However, the breakdown of nonbank-financing sources shows greater differences. Compared to other countries, firms in our sample rely on a large informal sector and alternative financing channels as suggested by Allen, Qian, and Qian (2005). These other financing sources could well be the large underground lending mechanism in China, which Farrell et al. (2006) note could represent up to a quarter of total bank deposits or several hundred billion dollars.

At the firm level, we find that, as expected, bank financing is more prevalent with larger firms.[8] We also find substantial firm level heterogeneity in financing patterns within China. The firms in the sample come from five different regions

[6] Besley and Levenson (1996) have also argued that economies like Taiwan (China) grew despite an underdeveloped formal financial sector, due to informal institutions.

[7] Other studies using the investment climate survey data on China include Dollar et al. (2004) and Cull and Xu (2005).

[8] The large firms in our sample are smaller compared to the listed firms in databases such as Worldscope used by most researchers. For instance, in our sample, the average large firm (in the top two quintiles of sales in 1999) had sales of 342 million Yuan or $41 million at the exchange rate of 8.2 Yuan/$. By comparison, Beck, Demirgüç-Kunt, and Maksimovic (2006) identify the average firm size of the one hundred largest firms in forty-four countries from Worldscope to be fifty-one times larger at $2.15 billion.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

of China: Coastal, Southwest, Central, Northwest, and Northeast. The financing patterns show that the largest amount of bank financing is in the Coastal (23.3%) and Southwest regions (26%), which have an investment climate that facilitates access to formal sources of external finance.

We find that firms using formal bank financing grow faster than those financed from alternative channels. Our results hold even when we exclude publicly traded companies or state-owned companies and look at a sample of just private sector firms, which are the fastest-growing firms in the Chinese economy. We also find that firms financed by banks experience higher reinvestment rates and have productivity growth at least equal to that of firms financed from nonbank sources. This suggests that the growth financed by banks is not inefficient growth.

The article's results do not rely on establishing causality, since even if it were the case that the fastest-growing firms are bank financed, by revealed preference, we know that at the margin banks are a better alternative to informal finance for fast-growing firms. However, we use a full battery of alternative specifications, a selection model, instrumental variables estimators, and a matching estimator to empirically test for the possibility that firms that obtain bank loans grow slower than similar firms that rely on informal or other sources of financing. We find that, controlling for the endogeneity of the bank financing decision, bank finance is associated with higher sales growth and profit reinvestment.

We also directly compare the performance of firms that report bank loans with that of firms that rely on external financing from informal sources. Firms that rely on informal external financing have lower profit reinvestment rates and do not grow faster or have higher productivity growth than firms that are bank financed. Only when informal financing is defined to include internal financing, as in Allen, Qian, and Qian (2005), is informal financing associated with higher productivity growth and reinvestment (but never with higher sales growth).

While we find the majority of firms that receive bank loans grow faster as a result, we do find a subpopulation of firms that do not. Firms that report that government help was instrumental in obtaining a bank loan, about 16% of the sample, do not show faster growth, higher reinvestment, or productivity compared to firms that get bank loans without government help. However, these results do not make China an exception to the growth and finance literature. They are consistent with the work on socialized banks by La Porta, Lopez-de-Silanes, and Shleifer (2002) and Cole (2009).

Overall, our results suggest that even in fast-growing economies like China, where the formal financial system serves a small portion of the private sector due to a poorly developed financial and legal system, external financing from the formal financial system is associated with faster growth and higher profit reinvestment rates for those firms that receive it. We find no evidence that alternative financing channels are associated with higher growth. While

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

we have some limited evidence that informal financing might help the smallest microsized firms, we find that all firms, irrespective of size, benefit from access to formal financing. Our findings suggest that the role of reputation and relationship-based informal financing and governance mechanisms in supporting the growth of private sector firms is likely to be limited and unlikely to substitute for formal mechanisms.

We discuss the Chinese economy and its financial system in Section 1. In Section 2, we describe the survey data and summary statistics, and in Section 3, we present our empirical model. Section 4 presents our regression results, Section 5 discusses the results, and Section 6 concludes.

1. The Chinese Financial System

1.1 The formal financial system

The formal financial sector in China consists of financial intermediaries, who are delegated monitors that operate with state charters. Within this formal sector, lending by state-owned banks dwarfs all other types of formal lending, such as equity, trade credit, leasing, or factoring, that make up a small percentage of GDP.

The banking sector is dominated by four state-owned banks that were commercialized in 1995.[9] Historically, a large share of bank funding has gone to state-controlled companies, leaving companies in the private sector to rely more heavily on alternative financing channels. The state banks have a high ratio of nonperforming loans to total loans (Cousin 2006 and annual statistics from the Chinese Banking Regulatory Commission), poor profitability, poor institutional framework, and weak corporate governance.[10]

Most bank loans in China are backed by collateral, and the only type of collateral acceptable to many banks is land or buildings (Gregory and Tenev 2001; Cousin 2006). Moveable assets are rarely used as security due to the highly restrictive secured financing lending system;[11] inventory and receivables cannot be used as collateral under Chinese Security Law (WB-PBOC Report,

[9] The four large state-owned banks include the Industrial and Commercial Bank of China (provides working capital loans to state industrial enterprises), China Agriculture Bank (specializes in agricultural lending), China Construction Bank (provides funds for construction and fixed asset investment), and the Bank of China (specializes in trade finance and foreign-exchange transactions). In addition to the big four state-owned commercial banks, there are also several minor players that include three policy (or development) banks (Export–Import Bank, Agricultural Development Bank, and State Development Bank), second-tier commercial banks (e.g., CITIC Industrial Bank), and regional banks such as housing savings banks, city banks, and other institutions such as Trust and Investment Corporations that capture smaller portions of the lending market. See Chen and Shih (2004) and Cousin (2006) for a detailed description of banking in China.

[10] This is not unique to China. Cole (2009) shows that government-owned banks in India have higher delinquent loan rates; La Porta, Lopez-de-Silanes, and Shleifer (2002) report that higher government ownership of banks is associated with slower growth across countries.

[11] According to the World Bank–People's Bank of China (2006) Report on "Secured Transactions Reform and Credit Market Development in China," only 4% of commercial loans are secured by movable assets.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

pp. 195–96).[12] The banks rely on courts and the state and local governments for repossessing collateral and to ensure loan repayment. The local governments in China have strong legislative powers recognized by the constitution and implement national laws according to their needs (Zhang and Booth 2001; Fan, Huang, and Zhu 2008).

Studies using macrodata report mixed findings on the association between financial development and growth in China. While Boyreau-Debray (2003) shows that banking depth is negatively correlated with interprovincial differences in growth rates, several other studies, including Liu and Li (2001), Zhou and Wang (2002), and Cheng and Degryse (2006), find that banking development is positively associated with growth across Chinese provinces. Cull and Xu (2000, 2003) focus only on the state sector in China using data on Chinese state-owned enterprises from 1980 to 1994 and find that even in the state sector, bank financing was associated with higher productivity, profitability, and adoption of reforms compared to government transfers. Thus, the inefficiency of Chinese bank lending should not be overstated, especially for recent years, since there has been a massive restructuring of the banking sector, including allowing entry of foreign banks, both to prepare the banks for equity listings and strategic sales and in anticipation of China's accession to the WTO. As argued in Cheng and Degryse (2006), Chinese banks have unique advantages, such as a larger geographic scope and fewer restrictions in attracting deposits, and are a less costly source of financing than other nonbank financing channels, such as urban and rural credit cooperatives.

While China's banking system is large, its equity and bond markets are smaller than that of most other countries, in terms of both market capitalization and total value traded as a percentage of GDP. The equity markets are largely a vehicle for privatization by the government rather than a market for capital raising by firms with growth opportunities, as shown in Wang, Xu, and Zhu (2004). The corporate bond market in China is crippled by excessive government regulation and the lack of institutional investors and credit rating agencies to help price the debt accurately. Durnev et al. (2004) show that compared to other transition economies, China has one of the poorest functioning stock markets, with highly synchronous stock returns that can be linked to weak property rights, corporate opacity, and political rent seeking. Other forms of formal financial institutions such as factoring and leasing are relatively underdeveloped in China (Gregory and Tenev 2001).[13]

[12] More recently, the People's Republic of China promulgated the Property Rights Law that went into effect on October 1, 2007, which extends the scope of security interests to include account receivables.

[13] According to statistics from the *World Leasing Yearbook*, while the total business volume of the global leasing industry topped $461.6 billion in 2003, China's share was very small, with only US$2.2 billion in business volume. And according to Klapper (2006), over 1994–2003, factoring as a percentage of GDP in China was only 0.05%.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

1.2 The informal financial system

The informal financial sector in China, also referred to as the nonstandard financial sector in Allen, Qian, and Qian (2008), consists of nondelegated monitors, such as pawnbrokers and moneylenders, as well as delegated lenders such as private money houses and informal banks that operate without state charters and are explicitly banned by the People's Bank of China.[14] As discussed below, the delegated informal lending institutions are illegal, although the enforcement of laws banning them by central and local government agencies is uneven and inconsistent over time. While it is difficult to estimate the size of the informal financial sector precisely, conservative estimates estimate it to be at least one-quarter of all financial transactions (Tsai 2002), with an estimated size of CNY 740–830 billion at the end of 2003 (Central Finance University of China, unpublished report on *Nationwide Survey on Informal Finance*.).

The informal lending institutions in China take various forms. Interpersonal lending (minjian jiedai) based on personal credit and reputation and trade credit (hangye xinyong) are largely legal and very commonly used among entrepreneurs. Our classification of trade credit as informal finance in China given its nondelegated nature is consistent with the existing literature on the Chinese informal financial system (e.g. Cull, Xu, and Zhu 2009). However, it must be noted that trade credit is regarded as formal finance in developed markets,[15] and in the absence of detailed data on trade credit lending technologies in China, we interpret this classification with caution. None of our main results on bank financing and informal financing are dependent on this classification. Pawnshops are legal in some areas and illegal in others and, as of 2000, were classified as a special kind of industrial and commercial enterprise regulated by the State Economic and Trade Commission. Rotating credit associations or *hui* (*hui* meaning community or association or club)[16] are organized by individuals, and while they are tolerated by the state in some provinces, they are shut down in other provinces and do not have access to formal legal channels (Tsai 2002, 2004). There are also private money houses and underground lending organizations that function like banks but charge very high interest rates above the state-mandated interest rate ceilings (Tsai 2002; Farrell et al. 2006) and are thus not sanctioned by the People's Bank of China.

---

[14] By contrast, in countries like the United States, the formal financial system consists of financial intermediaries functioning as delegated monitors, all of whom are legal. The informal system in these countries consists of nonintermediated finance from individuals. For instance, Berger and Udell (1998) differentiate between the venture capital market, which is formal intermediated finance, and the angel financing market, which is informal financing.

[15] Recent literature in finance regards trade credit as formal finance, and in some cases, a possibly more informed source of credit than bank finance. See Bias and Gollier (1997), Berger and Udell (2006), and Burkart, Ellingsen, and Giannetti (forthcoming) for a discussion of how trade credit shares features with both transactions- and relationship-based lending technologies.

[16] These associations take on various forms, including *biaohui* (where members tender for making use of their funds), *taihui* (which resemble pyramidal schemes), *lunhui* (where the turn of each member is predetermined), *yaohui* (where members rotate according to draws), and *bahui* (where money is divided according to group hierarchy) (Ma 2003).

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Access to the informal market is usually based on networking through common friends, and the cost of accessing informal credit varies depending on the structure of local social networks. While low-interest loans are common only among close friends and family (Tsai 2004), Allen, Qian, and Qian (2008) note that private moneylenders and money houses charge very high interest rates, with the associated loan contracts resembling junk bonds. The informal lenders rely on trust and reputation or coercion and violence for repayment of the loans and very often do not require collateral or guarantees. For instance, a survey by Zhang, Yuan, and Lin (2002) estimates that 84% of all informal lending in the Guangdong province is based on personal credit without collateral. The informal lenders that do require collateral are more lenient than banks in the types of collateral that they require, including oxen or personal assets such as motorcycles (Tsai 2002), and are willing to renegotiate loan terms and contracts. In addition, informal lenders rarely access the limited secured financing transactions market, if at all, and there are no separate clauses or articles in the Guarantee Law of China for informal lenders (Li 2005a,b; Han 2006; World Bank–People's Bank of China 2006). Thus, the enforcement mechanism of these informal institutions is not through the legal system or through government connections but through informal means, including violence as in the case of the illegal private money houses.[17]

1.3 Formal versus informal finance

To summarize the above discussion, the formal financial system in China operates with a state charter and consists of financial intermediaries as delegated monitors. The informal financial system consists of individual lenders or nondelegated monitors such as moneylenders and institutions that operate without state charters (Tsai 2004; Allen, Qian, and Qian 2008). This distinction between formal and informal financing channels manifests in the type of enforcement mechanisms used. Formal financial intermediaries such as banks typically lend conditional on collateral (Gregory and Tenev 2001; Cousin 2006) and rely on formal institutions such as courts or government channels to enforce repayment of loan contracts (Zhang and Booth 2001). By contrast, individual moneylenders and the informal delegated lenders that are deemed illegal by the People's Bank of China rarely require collateral and do not use the courts or the government and rely on informal channels to enforce repayment (Tsai 2004; OECD 2005).[18]

[17] The authors are grateful to Jianjun Li of the Central University of Finance and Economics, Beijing, China, for a personal illustration. When a relative borrowed RMB 300,000 from a lender, pledging his apartment (market value of RMB 600,000) as collateral, the lender's own relatives physically occupied the property until all the principal and interest was repaid.

[18] We find interesting differences in resolution mechanisms in our sample of firms as well. In univariate tests, we find that the percentage of client disputes resolved through court action and arbitration by the local government is significantly higher for firms with bank loans than for firms without bank loans. In multivariate regressions with all the usual controls for size, age, ownership, competition, and region, we find a strong and significant association between bank financing and resolution of client disputes through court action.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Our definition of formal versus informal financing in China overlaps with two commonly used definitions: the definition from the financial intermediation literature (e.g., Diamond 1984; Berger and Udell 1998) that informal financing is not associated with a delegated monitor, unlike formal financing,[19] and the definition in the economics literature (e.g., Kandori 1992; Udry 1994; Straub 2005) that informal financing is characterized by the use of self-enforcing contracts and social sanctions, including coercion and violence, to ensure repayment rather than formal legal enforcement mechanisms. This latter definition is consistent with the extensive theoretical literature in corporate finance on optimal financial structures, where investors can observe the firm's cash flows but cannot enforce legal rights to these cash flows (e.g., Hart and Moore 1995; Bolton and Scharfstein 1996). In the context of China, our definition of informal financing is also consistent with the definition of Allen, Qian, and Qian (2005) that informal financing is everything that is not bank financing, since the banking sector is the largest component of the formal financial sector. Allen, Qian, and Qian (2008) specifically refer to lending by the moneylenders and private money houses as the nonstandard financial sector.

2. Data and Summary Statistics

The data on Chinese firms come from the World Bank Investment Climate survey, which was undertaken in early 2003 in collaboration with the Enterprise Survey Organization of the Chinese National Bureau of Statistics (NBS). The Chinese survey is part of the World Bank Enterprise Surveys, which use standardized survey instruments and a uniform sampling methodology to benchmark the investment climate of countries across the world and to analyze firm performance. The Enterprise Surveys sample from the universe of registered businesses and follow a stratified random sampling methodology.

The firms are randomly surveyed from both manufacturing and services industries,[20] with a restriction on minimum firm size, where firm size is defined by number of employees. The minimum number of employees was set at twenty (fifteen) for manufacturing (services) firms. The sampling frame used in the World Bank survey closely matches the overall distribution of firms in China according to the statistics reported by the Chinese NBS. Small firms (less than three hundred employees) make up 72% of our sample, medium

[19] Berger and Udell (2006) also emphasize the need for distinguishing lending technologies based on several dimensions, including primary source of information, screening and underwriting policies/procedures, structure of loan contracts, and monitoring strategies and mechanisms. This definition of lending technologies is very useful in the context of developed countries. Uchida, Udell, and Yamori (2006) provide empirical validation of the lending technologies classification.

[20] The industries sampled include both manufacturing (apparel and leather goods, electronic equipment, electronic components, consumer products, vehicles, and vehicle parts) and services (accounting and related services, advertising and marketing, business logistics services, communication services, and information technology).

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

firms (three hundred to two thousand employees) make up 21.2%, and large firms (more than or equal to two thousand employees) make up 6.7%.

Twenty-four hundred firms were sampled from the following eighteen cities to represent the five main regions in China: (i) Northeast: Benxi, Dalian, Changchun, and Haerbin; (ii) Coastal: Hangzhou, Wenzhou, Shenzhen, and Jiangmen; (iii) Central: Nanchang, Zhenzhou, Wuhan, and Changsha; (iv) Southwest: Nanning, Guiyang, Chongqing, and Kunming; (v) Northwest: Xi'an and Langzhou.

The survey questionnaire has two main sections. The first section consists of questions on general information about the firm, its relations with clients and suppliers, its relations with the government, and qualitative questions asking for the manager's opinion on the business environment. The second section is based on interviews with the firm's accountant and personnel manager, asking for balance sheet information and other quantitative information on employee training, schooling, and wages. While most of the qualitative questions pertain only to the year 2002, a short panel from 1999 to 2002 is available for the quantitative questions.[21]

The survey allows us to identify firms on the basis of their registration status as corporations, state-owned companies, cooperatives, and other legal structures. In addition, each firm also provides a detailed breakdown of its ownership structure into domestic versus foreign owners as well as a more disaggregated breakdown into the percentage owned by individuals, managers, institutional investors, firms, and banks.

2.1 Financing patterns: China compared to other countries

Enterprise managers taking the survey were asked: "Please identify the contribution of each of the following sources of financing for your establishment's new investments (i.e., new land, buildings, machinery and equipment)—Local Commercial Banks (that includes state-owned commercial banks, other commercial banks, urban credit cooperatives, rural credit cooperatives); Foreign-Owned Commercial Banks; Trade Credit (supplier or customer credit); Investment Funds or Special Development Financing or Other State Services; Internal Funds or Retained Earnings; Loans from Family and Friends; Informal Sources (e.g., moneylender or informal bank); Equity (sale of stock to management, sale of stock to legal persons, public issue of marketable share to outside investors); and Other Sources." The sum of these proportions adds up to 100%. A similar breakdown of sources is available for

[21] In a pilot survey in 2001, firms were sampled from the following Chinese cities: Beijing, Tianjin, Shanghai, Guangzhou, and Chengdu. We do not use these cities in our article, since the time period over which the information is available is different (1997–2001 as opposed to 1999–2002). In addition, we have no information on profit reinvestment rates and only limited information on productivity. However, we have replicated our key results (the sales growth regressions in table 4) on this smaller sample, as well as on a sample combining firms from both the pilot survey and the main survey, and find our results to hold.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

working capital financing (i.e., inventories, accounts receivable, and cash).[22] For expositional ease, we use hereafter the term "financing" to mean "new investment financing" unless otherwise specifically noted. The survey also provides additional details on individual bank loans, including the year of access, collateral requirements, and duration and interest rate of the loan.

We adopt two different categorizations of the various sources of financing. In the first categorization, we have the following six groups: Bank Financing, which includes local commercial banks and foreign commercial banks; Informal Financing, which includes financing from informal sources such as a moneylender or an informal bank; Operations Financing, which includes trade credit; Equity Financing; Investment Funds, which includes investment funds or special development financing or other state services; and Internal Financing, which includes internal funds or retained earnings, loans from family and friends, and the other category.

In the second categorization, we adopt the Allen, Qian, and Qian (2005) classification of financing sources into two groups: Bank Financing, which includes local commercial banks and foreign commercial banks; and Self-Fund Raising, which includes all other sources, such as retained earnings, informal sources, loans from family and friends, trade credit, investment funds, equity, and the other category.[23] One limitation of our survey data is that the financing patterns are given in terms of proportions of financing, not as debt to asset ratios, as is common in previous literature.

In table 1, we compare firm-level financing patterns in China with other developing countries, for which data are obtained from other Investment Climate Surveys.[24] As of 2006, there were sixty-seven country surveys covering over forty thousand firms. Since the core survey is the same across all countries, we have comparable information on financing sources across the different countries. The only difference is that some surveys also have information on leasing arrangements and credit card financing, which is not provided in the case of China. We combine these two categories along with Trade Credit and label it Operations Financing in our tables.

[22] We find a strong correlation between the sources of new investments and sources of working capital. In our data, trade credit financing of new investments is a small proportion of overall financing. When we look at average statistics, trade credit financing of new investments was 1.03% and that of working capital was 2.27%. A t-test reveals the difference to be statistically significant at the 1% level.

[23] While the current literature considers trade credit, investment funds, and equity to be sources of external financing, we adopt this classification only to be consistent with the (2005) categorization. None of our regression results are dependent on classifying these sources as self-financing. Allen et al. also consider two additional financing sources that we do not have information on in our survey: the state budget and foreign investment. This is unlikely to influence our results on bank financing and informal financing, since we have only 116 firms in our sample that have more than 50% foreign ownership, and as Allen et al. mention, the state budget contributes to only 10% of state-owned companies' total funding.

[24] The Investment Climate Surveys are an ongoing initiative by the World Bank to study the investment climate conditions in developing countries and their impact on firm productivity, investment, and employment.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

In Panel A of table 1, we present individual financing patterns, and in Panel B, we present aggregate financing patterns. In both panels, we compare firm-level financing patterns (averaged across all firms) in China with those in other emerging markets, such as Bangladesh, Brazil, India,[25] Indonesia, Nigeria, and the Russian Federation. Panel A shows that China has the least amount of Internal Financing/Retained Earnings (15.24%) compared to all the other emerging markets. While the use of funds from Family and Friends and Informal Sources such as moneylenders or informal banks seems to be comparable with its use in other countries, China also has the highest average amount of financing from Other Sources (42.70%) compared to the other developing countries. The next largest dependence on financing from Other Sources is in Indonesia, but even there, only 8.8% of new investments are financed by Other Sources.

Several cross-checks show our data to be consistent with data used in other studies. For instance, OECD (2005) reports that the highest shares of informal finance are found in the northeast provinces of Heilongjiang and Liaoning, based on a nationwide survey of informal financing in twenty provinces (statistics on other provinces are not provided). When we sum the Other and Informal Sources in our data, we find that the Heilongjiang Province has the highest proportion of Other + Informal at 70%. Dalian, a city in the Liaoning Province, also has a high percentage of Other + Informal at 61.61%. Li (2005b) estimates that the total volume of informal lending in 2003 ranged from 740.5 billion RMB (US$91.42 billion) to 816.4 billion RMB (US$100.79 billion), which on average represents 28.07% of the total scale of lending by formal financial institutions. The World Bank–People's Bank of China (2006) survey of informal finance estimated the annual scale of informal lending to be 950 billion RMB (US$118 billion) or 6.96% of the country's GDP.[26] This is further confirmed by Tsai (2002, 2004) and Allen, Qian, and Qian (2005) and a recent McKinsey report (Farrell et al. 2006) that documents large underground lending in China amounting to several hundred billion US dollars.

While we do not have a detailed breakdown of what these other sources could be in our survey, we have evidence from a 2002 pilot survey (see Dollar et al. 2003) conducted in five cities in China prior to the survey used in this article. In the 2002 pilot survey, 107 firms report the exact composition of the Other category. None of the 107 firms reports Other to be a bank loan. The responses in the pilot survey indicate that in some cases, sources of funding such as income from land, government financing, and self-financing may have been classified as Other. The pilot survey also includes an additional category on financing sources, inter-firm transfers within a group, which is missing from the 2003 survey and we believe may have been misclassified as Other. As a re-

[25] The information on financing choices for Indian firms comes from the World Business Environment Survey, which was also conducted by the World Bank as a precursor to the Investment Climate Surveys. The Indian ICS does not have detailed information on firm financing choices.

[26] *Zhongguo jingying bao* [*China Management News*], August 6, 2005.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Table 1
Financing patterns in developing countries

Panel A: Individual financing patterns

	Number of countries	Number of firms	Retained Earnings	Local Commercial Banks	Foreign-Owned Commercial Banks	Operations Finance	Investment Funds	Loans from Family and Friends	Equity	Informal Sources	Other
Bangladesh		892	59.92	28.41	1.22	4.55	0.26	4.27	0.38	0.35	0.64
Brazil		1, 351	56.32	13.09	1.21	12.12	8.45	1.21	4.29	1.04	2.27
China		1, 342	15.24	20.24	0.12	1.03	0.55	5.89	12.39	1.84	42.70
India		92	43.84	30.73	2.75	0.43	9.52	3.56	4.33	0.75	4.09
Indonesia		291	41.89	13.21	3.13	5.49	1.67	17.73	1.34	6.74	8.80
Nigeria		145	63.94	29.76	0.00	1.07	1.55	0.74	2.59	0.34	0.00
Russian Federation		701	82.47	5.57	0.36	5.87	0.73	1.74	0.36	1.02	1.87
Across regions											
Africa	15	2, 708	66.02	18.14	1.05	6.15	0.98	1.88	1.36	0.48	3.95
East Asia and Pacific (excluding China)	8	5, 307	31.19	30.77	1.10	3.30	1.37	7.08	21.38	1.23	2.57
East Europe and Central Asia	36	15, 489	68.53	11.38	1.48	7.87	1.01	2.94	4.20	0.71	1.88
Latin America and Caribbean	10	4, 001	53.96	19.36	1.86	10.79	3.63	2.89	3.13	0.76	3.62
Middle East and North Africa	5	2, 103	74.52	12.37	0.69	5.25	0.28	2.46	1.70	0.15	2.57
South Asia	5	1, 572	58.63	21.95	0.86	4.76	1.00	4.59	3.23	0.68	4.29
Across income groups											
Low income	26	6, 913	59.22	16.32	1.10	3.23	1.30	5.47	10.43	0.94	1.99
Middle income	45	21, 062	59.26	16.12	1.40	7.29	1.43	3.26	4.89	0.82	5.53
High income OECD	7	3, 020	58.22	19.74	1.04	12.81	0.62	1.09	5.29	0.10	1.09

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 1
continued

Panel B: Aggregate financing patterns

	Number of countries	Number of firms	Aggregate financing patterns: Internal Financing	Bank Financing	Informal Financing	Operations Financing	Equity Financing	Investment Funds	Allen et al. (2005) categorization: Bank Financing	Self-Fund Raising
Bangladesh		892	64.84	29.64	0.35	4.55	0.38	0.26	29.64	70.36
Brazil		1, 351	59.80	14.30	1.04	12.12	4.29	8.45	14.30	85.70
China		1, 342	63.82	20.37	1.84	1.03	12.39	0.55	20.37	79.63
India		92	51.49	33.48	0.75	0.43	4.33	9.52	33.48	66.52
Indonesia		291	68.42	16.34	6.74	5.49	1.34	1.67	16.34	83.66
Nigeria		145	64.69	29.76	0.34	1.07	2.59	1.55	29.76	70.24
Russian Federation		701	86.08	5.93	1.02	5.87	0.36	0.73	5.93	94.07
Across regions										
Africa	15	2, 708	71.84	19.19	0.48	6.15	1.36	0.98	19.19	80.81
East Asia and Pacific (excluding China)	8	5, 307	40.84	31.87	1.23	3.30	21.38	1.37	31.87	68.13
East Europe and Central Asia	36	15, 489	73.34	12.86	0.71	7.87	4.20	1.01	12.86	87.14
Latin America and Caribbean	10	4, 001	60.47	21.22	0.76	10.79	3.13	3.63	21.22	78.78
Middle East and North Africa	5	2, 103	79.55	13.07	0.15	5.25	1.70	0.28	13.07	86.93
South Asia	5	1, 572	67.51	22.82	0.68	4.76	3.23	1.00	22.82	77.18
Across income groups										
Low income	26	6, 913	66.67	17.42	0.94	3.23	10.43	1.30	17.42	82.58
Middle income	45	21, 062	68.05	17.52	0.82	7.29	4.89	1.43	17.52	82.48
High income OECD	7	3, 020	60.40	20.78	0.10	12.81	5.29	0.62	20.78	79.22

This table presents financing patterns across developing countries, geographic regions, and country-income groups across the world. In Panel A, Retained Earnings, Local Commercial Banks, Foreign-Owned Commercial Banks, Equity, Operations Finance, Investment Funds, Loans from Family and Friends, Informal Sources (e.g., moneylender), and Other are financing proportions that stand for the proportion of new investments financed by each of these sources. Operations Finance consists of financing from leasing, trade credit, and credit cards. Investment Funds includes funds from investment funds, development bank, and other state services. In Panel B, we present two sets of aggregate financing patterns. The first set consists of Internal Financing (Retained Earnings, Loans from Family and Friends, and Other), Bank Financing (Local Commercial Banks and Foreign-Owned Commercial Banks), Informal Financing (Informal Sources such as moneylender or informal bank), Equity Financing, Operations Financing, and Investment Funds. The second set consists of Bank Financing and Self-Fund Raising (100−Bank Financing) as defined in Allen, Qian, and Qian (2005) to represent the proportion of new investments financed by all other sources except Bank Financing. The financing proportions are in percentages.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

sult, in our statistical analysis below, we follow Allen, Qian, and Qian's (2005) classification and rely primarily on the bank variable in our statistical tests.

China also looks unique in the large usage of Equity Financing (12.39%) compared to the other developing countries (all below 5%) in table 1. However, when we look across other geographic regions, the East Asia and Pacific region (excluding China) uses the largest amount of Equity Financing (21.38%) compared to other regions (below 5%).[27] The data allow for a more detailed breakdown of the equity issuance into sale of stock to employees, public issue of marketable shares to outside investors, and sale of stock to legal persons. The sale of stock to employees and public issuance of marketable shares to outside investors is quite low at 2.89% and 1.26%, respectively, and most of the equity issuance is really sale of stock to legal persons (8.24%). Legal person shareholders are unique to China and are analogous to institutional shareholders in western economies except that they tend to have strong links with the state[28] and are not widely held as in western economies.

In Panel B of table 1, we look at the aggregate financing patterns. When we look at Internal Financing, which includes retained earnings, loans from family/friends, and other sources, Chinese firms look comparable to firms in other countries. Focusing on the Allen, Qian, and Qian (2005) categorization, Chinese firms source 20% of their funds from banks and 80% from self-fund-raising channels. These numbers are very similar to the averages for countries in Africa (Bank Financing = 19%, Self-Fund-Raising = 81%), South Asia (Bank Financing = 23%, Self-Fund-Raising = 77%), and Latin America and the Caribbean (Bank Financing = 21%, Self-Fund-Raising = 79%). Countries in East Asia and the Pacific use slightly more Bank Financing (32%) and a lesser amount of Self-Fund-Raising (68%) compared to China, whereas countries in East Europe and Central Asia and those in Middle East and North Africa use less Bank Financing (13%) and more Self-Fund-Raising (87%) than even China.[29]

The financing numbers are consistent when we look across country-income groups. Use of Bank Financing ranges from 18% in Low and Middle Income

[27] A map of the Chinese cities where the survey was conducted and a breakdown of the financing patterns across cities are provided in the Supplementary Data in figures 1 and 2 and table A1.

[28] As described in Xu and Wang (1997) and Delios and Wu (2005), the legal person shareholder category comprises private companies, state-owned enterprises, and nonbank financial institutions, such as investment funds and security companies. The legal person shareholders differ from pure state shareholders in that they are profit seeking and have relatively more freedom than state shareholders in deciding firm strategy and profit allocation.

[29] When we recompute table 1 for three different size classes (one to nineteen employees, twenty to ninety-nine employees, and one hundred and over employees), Chinese firms in each size class use less internal financing and more "other" financing compared to similar-sized firms in the other emerging markets. However, they are similar in their use of bank financing compared to firms from other emerging markets in the same size class. When we look at aggregate financing patterns, small firms in China use 10.2% bank financing (region averages for small firms range from 6.4% in Europe and Central Asia [ECA] to 18.5% in Latin America and the Caribbean [LCR]), medium firms in China use 17.3% bank financing (region averages for medium firms range from 9.51% in Middle East and North Africa [MNA] to 33.55% in East Asia and Pacific [EAP] region, excluding China), and large firms in China use 28% bank financing (region averages for large firms range from 13.71% in ECA to 45% in EAP, excluding China).

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

countries to 21% in High Income countries, and use of Self-Fund-Raising is 79–83% across country-income groups. Overall, these figures suggest that Chinese firms are not an anomaly in their use of Self-Fund-Raising compared to other developing countries in contrast to the findings in Allen, Qian, and Qian (2005).

2.2 Financing patterns across firms in China

Table 2 shows how financing patterns vary across different types of firms in China. The five different regions vary considerably in their investment climate, that is, the overall institutional and policy environment that affects firms' competitiveness and growth. In their evaluation of investment climate across different regions in China, Dollar et al. (2004) consider the physical infrastructure, domestic entry and exit barriers, skills and technology endowment, labor market flexibility, international integration, private sector participation, tax burden, corruption, court efficiency, and access to finance. They show that the Eastern and Coastal areas (Yangtze and Pearl River deltas) are more developed, while inland cities especially in the West and the Northeast tend to have worse investment climates. Cities in the Central region appear to be in the middle in terms of their investment climate. The financing patterns seem to be correlated with the quality of investment climate, with the amount of Bank Financing being largest in the Southwest (26%) and Coastal (23%) regions and the smallest in the Northwest region (14%). We control for the variation in financing patterns across different cities in our regressions using city dummies.

Table 2 also shows that in financing capital expenditures, the very large firms use more Bank Financing (30%) than the microfirms and small firms (14–15%). Publicly listed companies finance a little over 33% of their new investments through Bank Financing and 67% from Self-Fund-Raising sources. By contrast, cooperatives fund only 18% of their new investments from banks and 82% from sources other than banks. When we look at ownership, domestic private firms use 20% Bank Financing and 80% Self-Fund-Raising compared to state firms that use 26% Bank Financing and 74% Self-Fund-Raising. In both cases, Other makes up a bulk of Self-Fund-Raising. Older firms (firms older than twenty years) use more of Bank Financing (27%) compared to firms less than five years old, which depend on Self-Fund-Raising for 80% of their financing needs.[30]

2.3 Access to formal bank finance

Panel A of table 3 presents the summary statistics for various measures that measure firms' access to the formal and informal financial systems in China.

[30] We obtain similar patterns, replacing financing sources of new investments with that of working capital.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Table 2
Financing patterns in China

Panel A: Individual financing patterns

	Number of firms	Retained Earnings	Local Commercial Banks	Foreign-Owned Commercial Banks	Operations Finance	Family and Friends	Informal Sources	Equity	Investment Funds	Other
Across regions in China										
Central	379	16.61	18.18	0.05	0.34	7.62	2.57	8.26	0.17	46.18
Coastal	265	15.27	23.19	0.47	1.46	3.39	0.34	16.49	0.49	38.89
Northeast	296	13.38	18.46	0.00	0.84	4.34	1.97	12.62	1.05	47.33
Northwest	89	13.07	14.15	0.00	1.12	6.29	1.78	11.42	0.80	51.38
Southwest	258	15.70	25.78	0.00	1.78	5.13	2.20	14.33	0.07	35.02
Across all firms	1, 287	15.16	20.52	0.11	1.03	5.41	1.85	12.39	0.46	43.07
Across firm sizes										
Micro	250	11.41	15.06	0.00	0.48	13.54	2.58	12.78	0.84	43.31
Small	249	14.20	14.82	0.04	1.14	6.70	1.59	13.97	0.10	47.45
Medium	249	16.50	21.89	0.46	0.60	4.62	1.49	11.14	0.94	42.36
Large	249	17.04	21.35	0.00	1.10	2.79	1.99	12.14	0.14	43.45
Very Large	249	17.02	30.13	0.08	1.61	0.08	1.81	11.25	0.29	37.73
Legal status										
Publicly listed company	37	9.00	33.32	0.00	0.00	0.00	0.00	36.11	0.00	21.57
Privately held company	490	18.11	20.54	0.03	0.98	10.00	2.21	14.14	0.26	33.75
Cooperatives/Collectives	165	15.92	17.59	0.00	0.55	5.33	0.99	10.65	0.06	48.92
Across firm ownership (>50%)										
State	288	11.69	25.90	0.00	0.02	0.79	1.47	6.36	0.59	53.18
Domestic private	901	16.23	19.60	0.14	1.10	7.30	2.00	14.16	0.36	39.12
Foreign private	108	16.19	14.56	0.19	3.36	1.67	1.39	13.64	0.93	48.08
Age										
Less than 5 years	262	14.83	20.53	0.00	0.73	10.29	2.61	16.33	0.52	34.18
Between 5 and 20 years	735	16.67	18.16	0.20	1.45	4.85	1.52	13.53	0.62	43.01
Greater than 20 years	290	11.67	26.51	0.00	0.24	2.40	1.99	5.96	0.00	51.23

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 2
continued

Panel B: Aggregate financing patterns

		Aggregate financing patterns						Allen et al. (2005) categorization	
	Number of firms	Internal Financing	Bank Financing	Informal Financing	Operations Financing	Equity Financing	Investment Funds	Bank Financing	Self-Fund Raising
Across regions in China									
Central	379	70.41	18.23		0.34	8.26	0.17	18.23	81.77
Coastal	265	57.55	23.66	0.34	1.46	16.49	0.49	23.66	76.34
Northeast	296	65.05	18.46	1.97	0.84	12.62	1.05	18.46	81.54
Northwest	89	70.74	14.15	1.78	1.12	11.42	0.80	14.15	85.85
Southwest	258	55.85	25.78	2.20	1.78	14.33	0.07	25.78	74.22
Across all firms	1287	63.64	20.63	1.85	1.03	12.39	0.46	20.63	79.37
Across firm sizes									
Micro	250	68.27	15.06	2.58	0.48	12.78	0.84	15.06	84.94
Small	249	68.35	14.86	1.59	1.14	13.97	0.10	14.86	85.14
Medium	249	63.47	22.35	1.49	0.60	11.14	0.94	22.35	77.65
Large	249	63.27	21.35	1.99	1.10	12.14	0.14	21.35	78.65
Very Large	249	54.84	30.21	1.81	1.61	11.25	0.29	30.21	69.79
Legal status									
Publicly listed company	37	30.57	33.32	0.00	0.00	36.11	0.00	33.32	66.68
Privately held company	490	61.86	20.57	2.21	0.98	14.14	0.26	20.57	79.43
Cooperatives	165	70.16	17.59	0.99	0.55	10.65	0.06	17.59	82.41
Across firm ownership									
State	288	65.66	25.90	1.47	0.02	6.36	0.59	25.90	74.10
Domestic private	901	62.65	19.74	2.00	1.10	14.16	0.36	19.74	80.26
Foreign private	108	65.94	14.75	1.39	3.36	13.64	0.93	14.75	85.25
Age									
Less than 5 years	262	59.29	20.53	2.61	0.73	6.36	0.59	20.53	79.47
Between 5 and 20 years	735	64.53	18.35	1.52	1.45	14.16	0.36	18.35	81.65
Greater than 20 years	290	65.30	26.51	1.99	0.24	13.64	0.93	26.51	73.49

This table presents financing patterns across different regions in China and across different types of firms. In Panel A, Retained Earnings, Local Commercial Banks, Foreign-Owned Commercial Banks, Equity, Operations Finance, Investment Funds, Loans from Family and Friends, Informal Sources (e.g., moneylender), and Other Sources are financing proportions that stand for the proportion of new investments financed by each of these sources. Operations Finance consists of financing from leasing, trade credit, and credit cards. Investment Funds includes funds from investment funds, development bank, and other state services. In Panel B, we present two sets of aggregate financing patterns. The first set consists of Internal Financing (Retained Earnings, Loans from Family and Friends, and Other Sources), Bank Financing (Local Commercial Banks and Foreign-Owned Commercial Banks), Informal Financing (informal sources such as moneylender or informal bank), Equity Financing, Operations Financing, and Investment Funds. The second set consists of Bank Financing and Self-Fund Raising (100−Bank Financing) as defined in Allen, Qian, and Qian (2005) to represent the proportion of new investments financed by all other sources except Bank Financing. The financing proportions are in percentages.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 3
Summary statistics and correlations

Panel A: Summary statistics

Variable	N	Mean	Standard deviation	Minimum	Maximum
Financing					
Bank Dummy	2, 326	0.2309	0.4215	0	1
Access Dummy	2, 400	0.2858	0.4519	0	1
Bank Financing	941	0.2721	0.4452	0	1
Self-Financing1	1, 446	0.6023	0.4896	0	1
Self-Financing2	1, 502	0.8302	0.3756	0	1
Firm Performance					
Sales Growth [2001–2002]	2370	0.0563	0.7072	−7.44	7.13
Reinvestment Rate [2001]	2, 115	0.1761	0.3230	0	1
Sales Growth [1999–2002]	2, 259	0.1349	0.3890	−2.36	2.71
Labor Productivity Growth [2001–2002]	1, 558	0.0045	0.8132	−6.68	8.02
Labor Productivity Growth [1999–2002]	1, 486	0.0810	0.3397	−1.97	3.28
Growth in TFP [2001–2002]	693	−0.0192	0.9521	−4.84	6.31
Firm Characteristics					
Size Dummies	2, 283	2.9991	1.4145	1	5
Age	2, 400	15.9862	14.3902	3	53
Corporation Dummy	2, 400	0.4046	0.4909	0	1
Cooperatives /Collectives Dummy	2, 400	0.1612	0.3678	0	1
State Ownership Dummy	2, 399	0.2213	0.4152	0	1
Competition Dummies	2, 326	3.8224	1.3535	1	5

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 3
Continued

Panel B: Correlations between financing and firm performance

	Sales Growth [2001–2002]	Reinvestment Rate	Labor Productivity Growth [2001–2002]	Sales Growth [1999–2002]	Labor Productivity Growth [1999–2002]	Bank Dummy	Access Dummy	Bank Financing	Self-Financing1
Reinvestment Rate	0.074***								
Labor Productivity Growth [2001–2002]	0.6536***	−0.0111							
Sales Growth [1999–2002]	0.5072***	0.1171***	0.2982***						
Labor Productivity Growth [1999–2002]	0.4238***	0.0488*	0.4682***	0.7298***					
Bank Dummy	0.037*	0.132***	0.0058	0.0671***	0.0573**				
Access Dummy	0.0003	−0.0076	−0.0332	−0.0613***	−0.0601**	0.0914***			
Bank Financing	0.0366	0.1664***	0.0279	0.074***	0.0878***	1***	0.0767***		
Self-Financing1	−0.0077	−0.0923***	0.0639**	−0.0491**	0.0016	−0.1174***	−0.0854***	−0.1762***	
Self-Financing2	0.0009	−0.0672***	0.0746***	−0.0229	0.0111	−0.1281***	−0.0801***	−0.1861***	0.9303***

*Significant at 10%; **significant at 5%; ***significant at 1%.

Panel A presents summary statistics. Panel B presents the correlations between the financing variables and firm performance. Bank Dummy takes the value of 1 if the firm states that it has a current loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm said it had no bank loan and had no overdraft facility or line of credit. Access Dummy is a dummy variable that takes the value of 1 if the firm had access to a bank loan in any year prior from 1990–2001 and 0 otherwise. Bank Financing Dummy takes the value of 1 if the firm states that it has a loan from a bank or financial institution and an overdraft facility or line of credit and reports that the bank finances at least 50% of new investments or working capital. Bank Financing takes the value of 0 if the firm states that it has no loan from a bank or financial institution and said it had no overdraft facility or line of credit and bank financing of both new investments and working capital was equal to 0%. Self-Financing1 takes the value of 1 if the sum of Informal Financing and Other Financing of either new investments or working capital is greater than 50%. Self-Financing1 takes the value of 0 if the sum of informal and other Financing of new investments and working capital is equal to 0%. Self-Financing2 takes the value of 1 if the sum of Internal, Informal, Family, and Other financing of new investments or working capital is greater than 50%. Self-Financing2 takes the value of 0 if the sum of internal, informal, family, and other financing of new investments and working capital is equal to 0%. Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002 and from 1999 to 2002. Labor Productivity Growth is defined as the log change in labor productivity where labor productivity is (Sales − Total Material Costs)/Total Number of Workers. Productivity Growth is computed from 2001 to 2002 and from 1999 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Firm Size Dummies are quintiles of total firm sales in 1999. Age is the age of the company in 2003. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperatives/Collectives Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. Competition takes values 1 to 5 for 1–3 competitors, 4–6 competitors, 7–15 competitors, 16–100 competitors, and over 100 competitors, respectively, for the number of competitors in its main business line in the domestic market.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

As the main independent variable of interest, we have an indicator variable, *Bank Dummy,* showing whether the firm has access to the formal financial system. *Bank Dummy* takes the value of 1 if the firm states that it has a bank overdraft facility or line of credit (L/C) and that it currently has a loan from the bank. The dummy takes a value of 0 if the firm states that it has no bank loan and no overdraft facility or line of credit from the bank. The *Bank Dummy* variable captures the use of bank financing. According to Berger and Udell (1995), bank L/Cs represent bank–borrower relationships and are relationship driven compared to other bank loans, which may be more transaction based. In our sample, having a current bank loan is the same as having a bank L/C, because there are no firms that report having a bank loan but no L/C. Further, only forty-nine firms report having access to an L/C but no current bank loan. For these firms, *Bank Dummy* = 0.[31] While we are unable to conduct formal tests of the difference between lines of credit and bank loans, we interpret our bank variables to be in the spirit of relationship lending. We also construct an *Access Dummy*, a dummy variable that takes the value of 1 if the firm had a past bank loan in any year prior, from 1990 to 2001, and 0 otherwise.

We also consider alternative indicators of access based on the financing proportions of new investments and working capital that firms report. The *Bank Financing Dummy* takes the value of 1 only if the firm states that it has a loan from a bank or financial institution and an overdraft facility or line of credit and reports that a bank finances at least 50% of new investments or working capital. *Bank Financing* takes the value of 0 if the firm states that it has no loan from a bank or financial institution and said it had no overdraft facility or line of credit and the bank financing of both new investments and working capital was equal to 0%. Thus, a firm is assigned a value of 1 for *Bank Financing* only if we have evidence of substantial reliance on bank financing and a value of 0 if it does not rely on bank financing.

We employ two measures of self-financing. *Self-Financing1* takes the value of 1 if the sum of financing of either new investments or working capital from Informal and Other Sources is greater than 50%.[32] *Self-Financing1* takes the value of 0 if the sum of financing of new investments and working capital from Informal and Other Sources is equal to 0%. *Self-Financing2* broadens the

[31] Redefining the *Bank Dummy* variable to take the value of 1 for the forty-nine firms does not change our results. In other robustness checks, we find that excluding these firms does not make a material difference to the results. In addition, we would expect these firms to be cash-rich firms with good opportunities. Raw correlations over this small sample of firms show that only internal financing is positively associated with sales growth, whereas alternate financing channels such as informal financing and other financing are not positively associated with sales growth.

[32] While the 50% hurdle rate for bank financing of working capital may be too high in the case of developed countries like the United States, bank financing of working capital is quite high in developing countries, consistent with the finding in Demirgüç-Kunt and Maksimovic (1999) that firms in developing countries have less long-term debt and more short-term financing. For instance, data from other World Bank Investment Climate Surveys reveal that average bank financing of working capital in Bangladesh is 32.15%, Brazil is 25.53%, Indonesia is 13.05%, and Chile is 26%. In our survey data, average bank financing of working capital in China is 26.26%.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

definition of self-financing and takes the value of 1 if the sum of financing of new investments or working capital from Retained Earnings, Loans from Family and Friends, Informal Sources, and Other Sources is greater than 50%. We combine Retained Earnings with external informal sources in *Self-Financing2* only to be consistent with Allen, Qian, and Qian (2005).

In most of our regressions, we focus on the *Bank Dummy* variable, since it is clearly defined and is not subject to any misclassification error and is not affected by the definition of the Other financing category. The summary statistics for the three bank variables, *Bank Dummy*, *Access Dummy*, and *Bank Financing Dummy*, indicate that on average, only 23–28% of the firms in the sample have access to bank financing. For instance, only 537 out of the 2,326 firms answering the bank loan question report having a bank loan. Of the full sample of twenty-four hundred firms, 1,466 firms report one of two reasons for not having an existing loan: 1,237 firms report not applying for a bank loan, and 229 firms report not having a bank loan, because their application for a bank loan was rejected. When we look at firms that are not registered as publicly traded corporations or as state-owned enterprises, of the 1,666 private firms, 1,301 firms report not having an existing loan from a bank or a financial institution, 933 private firms report not having a bank loan, because they did not apply for the loan, and 154 private firms report not having a bank loan, because their application for a bank loan was rejected.

The firms that report their loan application was rejected report three mutually exclusive reasons for why their application was rejected: lack of collateral, perceived lack of feasibility of project, and incompleteness of application. Of the 229 firms that report their loan application being rejected, 66% (152 firms) report lack of collateral as the main reason why their loan was rejected. This includes 104 private sector firms out of a sample of 154 private sector firms that report their loan application was rejected.

Thus, we are able to identify firms that may or may not need bank financing but do not apply for loans, as well as firms that apply for loans but have their loan applications rejected. The summary statistics also show collateral to be the main constraint for access to bank loans.

2.4 Firm performance and control variables

Our principal measure of firm performance is *Sales Growth*, which is computed as the log change in firm sales over the period 2001–2002. We supplement *Sales Growth* with two additional indicators: *Labor Productivity Growth* over the period 2001–2002 and the firm's *Reinvestment Rate* in 2002. *Labor Productivity Growth* is the log change in labor productivity, where labor productivity is defined as (Sales − Total Material Costs)/Total Number of Workers. *Productivity Growth* shows whether a source of financing is associated with declines in firm efficiency. *Reinvestment Rate* is the manager's estimate

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

of the percentage of net profits that are reinvested in the establishment and not distributed to owners, the state, or the shareholders. The firm's reinvestment rate shows whether the firm's managers are committing the firm's resources to finance growth or whether the firm's owners are using bank financing to extract resources from the firm. The latter case would be suggestive of inefficient investment.[33]

We also consider firm performance over a longer horizon by looking at *Sales Growth* and *Productivity Growth* over the period 1999 to 2002. As a robustness check, we also compute *Growth in Total Factor Productivity* (*TFP*) as the log change in TFP between 2001 and 2002. *TFP* is computed as the residual from a regression of Value Added on Labor and Fixed Assets, both interacted with industry sector dummies. Value Added is defined as Sales − Total Material Costs, Labor is the total number of employees, and Fixed Assets serves as a proxy for capital. *Growth in TFP* is not the main productivity growth measure in all our regressions, since the sample size is reduced by more than half with this measure.

The mean *Sales Growth* and *Labor Productivity Growth* in the full sample from 2001 to 2002 are 5.6% and 0.45%, respectively. The corresponding figures for the period 1999 to 2002 are 13.5% and 8.1%, respectively. The mean *Growth in TFP* over the period 2001–2002 is −1.92%.

In addition to the rich detail on the financing choices of firms, the survey has information on firm size, age, ownership, legal organization, and capacity utilization, all of which are used as firm-level controls in our study. An important strength of the survey is its wide coverage of micro- and small-sized firms.[34] We construct *Size Dummies*, which are the quintiles of firm's sales in 1999. The survey thus provides data across a much broader cross-section of firm sizes than is available in commercial databases, such as Worldscope.

Panel A of table 3 shows that the average firm *Age* in the sample is sixteen years. We include dummy variables to identify very young firms (one to five years), middle-aged firms (five to twenty years), and older firms (older than twenty years). We also use dummy variables to classify firms in terms of their current legal status into *Corporations*, *Cooperatives/Collectives*, and an *Other Legal Status* category. Corporations, both public and private, make up 40% of our sample, whereas 16% of our sample is composed of cooperatives or collectives. We also have detailed information on the ownership breakdown of these companies and the percentage owned by different entities in the private sector (e.g., domestic firms, domestic institutional investors, foreign individuals, foreign firms, foreign institutional investors, etc.) and different

[33] We do not use profitability as a measure of firm performance, since it is subject to measurement error. However, all our key results in table 4 are robust to using earnings before interest and taxes in 2002 as a dependent variable.

[34] The mean (median) number of employees in a firm in our sample in 1999 was 579 (one hundred) employees with 33% of our sample composed of firms with less than fifty employees.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

entities in the state sector (e.g., national government, state/provincial government, local/municipal government or other government bodies like collectives, etc.). We use a dummy variable, *State Ownership*, to characterize firms where the government owns more than 50% of the company. Nearly 22% of the sample (531 firms) is composed of firms with more than 50% state ownership, and the remaining sample is made up of firms with more than 50% private ownership. We also identify the number of competitors of the firm in its main business line in the domestic market by using five *Competition* dummies for one to three competitors, four to six competitors, seven to fifteen competitors, sixteen to one hundred competitors, and over one hundred competitors, respectively.

Panel B of table 3 presents the correlation between the access variables and firm performance. We find that the bank finance variables are all significantly correlated with each other; the *Bank Dummy* and the *Bank Financing Dummy*, where the latter variable takes into account bank financing of new investments and working capital, are perfectly correlated. *Bank Financing* and *Bank Dummy* are also positively correlated with the firm performance measures, whereas the *Access Dummy*, which focuses on access to past bank financing, appears to be negatively correlated. Both the self-financing dummies are negatively correlated with all measures of bank finance. They also appear to be negatively correlated with *Sales Growth* and *Reinvestment Rate* and have a positive association with *Productivity Growth*.

The raw correlations between firm performance and the financing variables mask some of the underlying variation. The average 2001–2002 growth rate of firms that receive bank financing is 10.34%, compared to an average growth rate of 4.2% for firms that receive no bank financing. The differences are starker when we look at the growth and financing figures by size quintiles. The lowest quintile of firms (with an average of only eighteen employees) has average growth rates of 10%, but only 6% of these firms have bank loans. These firms, however, are typical of the fast-growing microsized firms in most economies, which depend mainly on internal and informal sources for their growth. When we look at the second to fifth quintiles of firms with the number of employees ranging from forty-four to 2,475, we see a positive association between bank financing and growth. The proportion of bank loans varies from 16% in the second quintile to 41% in the fifth quintile, with the corresponding sales growth figures ranging from 0.02% to 6.9%.[35]

[35] The numbers are similar when we define size in terms of total sales. The lowest quintile of firms with average sales around 384,000 RMB has the highest growth rates of 12% and the least amount of bank loans (9%). The second to the fifth quintile of firms range in sales from 2,005,000 RMB to 414,353,000 RMB, with monotonic increases in the proportion of bank loans and sales growth with size. The proportion of bank loans ranges from 13% to 45%, and sales growth ranges from 1.3% to 4.8% as we move from the second quintile to the fifth quintile of firms.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

3. Financing Patterns and Firm Performance

We first estimate the following regression model:

$$\begin{aligned} Sales\ Growth/Reinvestment\ Rate/Productivity\ Growth = \\ \alpha\ &+\beta_1 Bank\ Dummy + \beta_2\ Size\ Dummies \\ &+\beta_3 Age\ Dummies + \beta_4 Corporation\ Dummy \\ &+\beta_5 Collectives\ Dummy + \beta_6 State\ Ownership\ Dummy \\ &+\beta_7 Competition\ Dummies + \beta_8 City\ Dummies + \varepsilon. \end{aligned} \tag{1}$$

The bank loans for our sample were approved prior to 2002. Accordingly, we report regressions for which the dependent variable is sales or productivity growth between 2001 and 2002 or the reinvestment rate measured in 2002. However, because annual growth rates are likely to be subject to random shocks, we also report regressions using sales growth and productivity growth of firms for the period 1999–2002. These regressions are descriptive, showing the association between growth and firm access to formal finance over a longer horizon.

Our main independent variable of interest is *Bank Dummy*. We include a number of firm-level control variables—dummies for size, age, state ownership, legal status, and competition—that are described in detail in Section 2.4. We include city dummies to control for unobserved heterogeneity at the city level. The data on industries are available for a very narrow definition of business activity, giving us eighty-one industry dummies. Hence, while our results are robust to controlling for industry dummies, we do not include industry dummies in our baseline specification so as to not lose too many degrees of freedom. We also report alternative specifications using the other financing variables described in Section 2.3—*Access Dummy*, *Bank Financing Dummy*, and the two informal financing variables, *Self-Financing1* and *Self-Financing2*.

3.1 Endogeneity of the bank financing decision

While Equation (1) establishes a broad association between formal and informal financial systems and firm performance, a positive relation observed between external finance and firm growth (or a negative one between self-fundraising and firm growth) may be simply due to reverse causality. To the extent that we are only interested in establishing an association between the use of bank financing and high growth rates, the direction of causality is immaterial. Even if it were the case that the fastest-growing firms go to banks for financing, then by revealed preference, bank financing is the preferred alternative to informal financing for such firms.

To investigate causality, we need to take into account that the bank financing decision is endogenous, and firms self-select into applying for bank loans; hence, we need to control for selection effects. We are interested in the effect

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

of bank financing on the performance of bank-financed firms (i.e., the difference between the performance of bank-financed firms and their performance if they were not bank financed). This difference is referred to as the Average Treatment Effect on the Treated (ATT). If Y_{i1} is the performance of the firm if it is bank financed and Y_{i0} is the performance if it is not bank financed, the ATT is given by:

$$\tau|_{\mathbf{Bank=1}} = \mathbf{E}(Y_{i1}|\ Bank_i = 1) - \mathbf{E}(Y_{i0}|\ Bank_i = 1). \tag{2}$$

Since we cannot observe the performance of bank-financed firms if they were not bank financed (i.e., $E(Y_{i0}|\ Bank_i{=}1)$), we estimate the following equation:

$$\tau = E(Y_{i1}|\ Bank_i = 1) - E(Y_{i0}|\ Bank_i = 0). \tag{3}$$

Equation (3) is a biased predictor of Equation (2), since the provision of bank financing is not randomly determined. The Heckman (1979) two-stage model explicitly addresses bias caused by correlation of the regressor with omitted variables by adding the inverse Mills ratio, which represents the nonzero expectation of the error term. A common interpretation of this term is to consider it as private information driving the selection decision. To estimate the selection model using instrumental variables, we need an instrument that is correlated with bank financing at the firm level, yet uncorrelated with firms' growth opportunities.[36]

As discussed in Section 1.1, one of the factors that affects firms' access to bank loans in China is the ability to post collateral in the form of land or buildings (Gregory and Tenev 2001; Tsai 2002; Cousin 2006). In our survey, firms were asked to report what share of collateral was Land and Buildings, Machinery, Intangible Assets (accounts receivable and inventory), and Personal Assets (e.g., house). As expected, most of the collateral posted was in the form of land or buildings (mean = 63.11%, median = 80%), followed by machinery (mean = 17.87%, median = 0%), personal assets (mean = 10.30%, median = 0%), and intangible assets (mean = 4.89%, median = 0%). Firms report that the main reason why banks reject their loan applications is their inability to meet collateral requirements.[37] In contrast, collateral plays a far less important role for loans from informal financiers, as discussed in Section 1.2.

Thus, as an instrument for bank financing, we construct a dummy variable *Collateral,* which takes the value of 1 if the firm reported that the financing required collateral and takes the value of 0 if the firm reported that the financing

[36] See Li and Prabhala (2007) for a discussion of selection models in corporate finance.

[37] More details are provided in Figure 3 of the Supplementary Data. The OECD (2005) also reports data from other surveys that find that more than 50% of the firms surveyed reported their lack of collateral as a major barrier to bank borrowing. The report further states that collateral or loan guarantees, or both, have become an essential precondition for most SME lending in China. Consistent with this, Li (2005a, b) also shows that more than half of SME queried in Zhejiang province, where the private sector is relatively highly developed, cited lack of collateral as their chief difficulty in obtaining bank loans.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

did not need collateral or that it did not apply for a loan because of stringent collateral requirements or that it was rejected for a loan because of the lack of collateral. Thus, *Collateral* serves as a proxy for the firm's ability to post collateral.[38] Ability to post collateral has been used as an instrument for bank financing in prior work, including Johnson, McMillan, and Woodruff (1999).

In our estimates, we allow for the possibility that the selection of firms receiving bank loans may be caused by firm characteristics unobserved by us but observable by banks. Specifically, we assume that a firm obtains access to formal bank finance (i.e., *Bank Dummy* = 1) if it meets the formal criteria of the banking system, so that the linear function of information we observe and the proprietary information observed by the bank exceed a threshold. Therefore, *Bank Dummy* = 1 if:

$$\begin{aligned}\alpha_0 \;\; &+\beta_1 Collateral + \beta_2 Size\ Dummies + \beta_3 Age\ Dummies \\ &+\beta_4 Corporations + \beta_5 Collectives + \beta_6 State\ Ownership \\ &+\beta_7 Competition\ Dummies + \beta_8 City\ Dummies + \zeta > 0, \end{aligned} \tag{4}$$

where $\zeta \sim (0,\sigma^2)$ is proprietary information observed by the bank. Equation (4) is also referred to as the Selection or Treatment Equation and forms the first stage of a two-stage selection model in which Equation (5) forms the second stage:

$$\begin{aligned}&Sales\ Growth/Reinvestment\ Rate/Productivity\ Growth = \\ &\quad\alpha_1 \;\; +\gamma_1 BankDummy + \gamma_2 Size\ Dummies + \gamma_3 Age\ Dummies \\ &\quad\quad +\gamma_4 Corporations + \gamma_5 Collectives + \gamma_6 State\ Ownership \\ &\quad\quad +\gamma_7 Competition\ Dummies + \gamma_8 City\ Dummies + \lambda + \varepsilon. \end{aligned} \tag{5}$$

The existence of a collateral requirement for a loan does not, by itself, affect a firm's growth and hence need not enter the second stage.[39] Thus, collateral serves as the identifying variable in our selection equation. We first obtain estimates of the selection equation and from these estimates compute the nonselection hazard λ (inverse of the Mills ratio) for each observation. λ is an estimate of the banks' private information underlying the selection of firms. The regression Equation (5) is then augmented with the estimate of the selection bias, the nonselection hazard, λ.

[38] We find that size and possession of fixed assets are determinants of a firm's ability to post collateral. In addition, we find that firms in cities with poor institutional environment but with higher fixed assets post more collateral. This suggests that in poor institutional environments, firms have to rely on collateral to access bank finance rather than relying on credit histories and growth opportunities.

[39] When we include the collateral requirement dummy in the second stage, it is not statistically significant.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

4. Results

4.1 Bank finance and firm performance

Table 4 reports the estimated coefficients from baseline regression (1). Columns (1) and (7) show that the formal financial system is associated with higher sales growth for both the full sample of all firms and a subsample of private firms (dropping public and state-owned firms). The role of bank financing in the second subsample is of particular interest, since private firms are likely to have fewer alternative options than publicly traded firms or state-owned firms. Firms that have bank financing also reinvest a higher proportion of their profits in their businesses in both samples (columns (2) and (8)). Thus, there is no evidence that bank financing funds growth that firms are unwilling to undertake using their own capital.

Columns (3) and (9) show that bank finance is not significantly negatively associated with labor productivity growth. When we look at growth in TFP for a smaller sample of firms in columns (4) and (10), we find that bank financing is positively associated with growth in TFP though not significantly so. These findings suggest that formal financial channels rather than informal sources have a positive association with firm growth and reinvestment and that the growth that results is not inefficient. Columns (5) and (6) for the full sample and columns (11) and (12) for the sub-sample reinforce this finding by showing that the association holds over a longer horizon. When we look at the control variables, larger firms, older firms, and firms organized as cooperatives or collectives are found to have lower growth rates. Growth rates are lower in highly competitive (more than one hundred competitors) industries. Our results are robust to controlling for fourteen industry dummies.

As robustness, in Panel B of table 4, we show whether the proportion of bank financing affects the outcomes we measure. We examine how new investments financed by bank loans and working capital financed by bank loans affect our measured outcomes, separately. This breakdown allows us to see whether, in our sample, bank financing of working capital may be more important in facilitating a firm's growth than bank financing of fixed assets, as suggested in the U.S. context by Berger and Udell (1995).

However, the relation between firm performance and the proportion of external financing may not be monotonic, since a heavy dependence on external financing of working capital may reflect financial distress. In addition, the continuous versions of the proportion variables in our data have a highly skewed distribution with a large point mass at 100%. To address these issues, we also construct a dummy variable, which takes the value of 1 when bank financing of new investment equals 100% and 0 otherwise (columns (1)–(6)), and a dummy variable, which takes the value of 1 when bank financing of working capital equals 100% and 0 otherwise (columns (7)–(12)).

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Table 4
Bank financing and firm performance—partial correlations

Panel A: Bank Dummy

	Full sample						Drop public corporations and state-owned companies					
	1	2	3	4	5	6	7	8	9	10	11	12
	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Growth in TFP [2001–2002]	Sales Growth [1999–2002]	Productivity Growth [1999–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Growth in TFP [2001–2002]	Sales Growth [1999–2002]	Productivity Growth [1999–2002]
Bank Dummy	0.075**	0.078***	−0.002	0.052	0.115***	0.058***	0.068*	0.086***	−0.004	0.051	0.120***	0.066**
	[0.034]	[0.020]	[0.051]	[0.084]	[0.019]	[0.022]	[0.040]	[0.024]	[0.057]	[0.098]	[0.024]	[0.026]
Small	−0.121**	0.057**	−0.064	−0.051	−0.195***	−0.085**	−0.127**	0.077***	−0.09	−0.061	−0.205***	−0.097**
	[0.057]	[0.023]	[0.079]	[0.253]	[0.031]	[0.038]	[0.064]	[0.026]	[0.084]	[0.258]	[0.034]	[0.043]
Medium	−0.136**	0.071***	0.007	−0.138	−0.258***	−0.136***	−0.146**	0.086***	−0.007	−0.186	−0.289***	−0.168***
	[0.056]	[0.024]	[0.080]	[0.239]	[0.031]	[0.038]	[0.065]	[0.028]	[0.088]	[0.247]	[0.035]	[0.043]
Large	−0.160***	0.047**	−0.088	−0.357	−0.298***	−0.187***	−0.169***	0.058**	−0.123	−0.347	−0.319***	−0.202***
	[0.057]	[0.023]	[0.081]	[0.247]	[0.032]	[0.040]	[0.066]	[0.027]	[0.089]	[0.252]	[0.037]	[0.046]
Very Large	−0.230***	0.075***	−0.086	−0.431*	−0.359***	−0.184***	−0.221**	0.108***	−0.086	−0.343	−0.398***	−0.220***
	[0.072]	[0.026]	[0.087]	[0.250]	[0.036]	[0.043]	[0.094]	[0.034]	[0.101]	[0.257]	[0.044]	[0.053]
Mid-Age	0.001	−0.007	−0.038	−0.113	−0.143***	−0.090***	−0.006	−0.027	−0.04	−0.175	−0.163***	−0.098***
	[0.041]	[0.021]	[0.059]	[0.104]	[0.024]	[0.029]	[0.047]	[0.024]	[0.063]	[0.118]	[0.026]	[0.032]
Old	−0.064	−0.048**	0.03	−0.091	−0.208***	−0.046	−0.035	−0.086***	0.064	−0.086	−0.210***	−0.06
	[0.049]	[0.024]	[0.075]	[0.146]	[0.027]	[0.035]	[0.064]	[0.032]	[0.083]	[0.195]	[0.035]	[0.040]
Corporations	−0.03	0.083***	−0.027	−0.097	−0.012	−0.038*	−0.066	0.106***	−0.047	−0.096	−0.043**	−0.041
	[0.035]	[0.019]	[0.057]	[0.086]	[0.018]	[0.022]	[0.043]	[0.022]	[0.069]	[0.099]	[0.021]	[0.026]
Cooperatives/ Collectives	−0.116**	0.033	−0.024	0.011	−0.143***	−0.098***	−0.169***	0.057**	−0.05	0.006	−0.184***	−0.098***
	[0.053]	[0.022]	[0.068]	[0.128]	[0.025]	[0.027]	[0.065]	[0.028]	[0.081]	[0.140]	[0.030]	[0.032]
State	−0.01	−0.022	0.053	−0.024	−0.037*	−0.033	−0.002	−0.02	−0.019	0.149	0.01	−0.019
	[0.041]	[0.019]	[0.062]	[0.162]	[0.020]	[0.026]	[0.076]	[0.035]	[0.115]	[0.296]	[0.039]	[0.046]
4–6 Competitors	0.012	0.053	−0.049	−0.181	0.011	−0.029	−0.017	0.092**	−0.004	−0.116	0.021	−0.02
	[0.055]	[0.033]	[0.086]	[0.157]	[0.033]	[0.038]	[0.070]	[0.039]	[0.104]	[0.191]	[0.043]	[0.047]
7–15 Competitors	0.015	0.004	0.011	0.055	−0.031	−0.03	0.019	0.033	0.075	0.003	−0.007	0
	[0.062]	[0.029]	[0.097]	[0.169]	[0.034]	[0.042]	[0.080]	[0.035]	[0.126]	[0.199]	[0.043]	[0.053]
16–100 Competitors	−0.053	0.022	−0.043	−0.053	−0.063**	−0.077**	−0.051	0.063*	−0.023	−0.032	−0.06	−0.074*
	[0.049]	[0.027]	[0.078]	[0.157]	[0.029]	[0.036]	[0.063]	[0.032]	[0.096]	[0.198]	[0.038]	[0.045]
>100 Competitors	−0.125**	0.02	−0.073	−0.052	−0.091***	−0.081**	−0.107*	0.046	−0.029	0.012	−0.083**	−0.073*
	[0.050]	[0.025]	[0.075]	[0.141]	[0.027]	[0.034]	[0.061]	[0.029]	[0.091]	[0.174]	[0.036]	[0.043]

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 4
Continued

Constant	0.288***	−0.022	0.151	0.231	0.489***	0.349***	0.360***	−0.071	0.246	0.283	0.519***	0.319***
	[0.099]	[0.036]	[0.144]	[0.281]	[0.052]	[0.061]	[0.136]	[0.048]	[0.166]	[0.301]	[0.067]	[0.074]
Observations	2,145	1,905	1,456	643	2,135	1,423	1,534	1,363	1,099	519	1,527	1,072
R-squared	0.036	0.072	0.017	0.054	0.175	0.071	0.037	0.087	0.023	0.049	0.195	0.083

Panel B: Alternate definitions of bank financing

	1	2	3	4	5	6	7	8	9	10	11	12
	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Growth in TFP [2001–2002]	Sales Growth [1999–2002]	Productivity Growth [1999–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Growth in TFP [2001–2002]	Sales Growth [1999–2002]	Productivity Growth [1999–2002]
Dummy for Bank Financing of New Investments=100%	−0.177	−0.135*	−0.164	−0.435	−0.205***	−0.149						
	(0.125)	(0.0752)	(0.204)	(0.325)	(0.0704)	(0.0932)						
Bank Financing of New Investments (Continuous Variable)	0.199*	0.101	0.103	0.105	0.235***	0.174**						
	(0.107)	(0.0680)	(0.185)	(0.292)	(0.0696)	(0.0879)						
Dummy for Bank Financing of Working Capital=100%							−0.127*	−0.190***	−0.156	−0.327	−0.176***	−0.0679
							(0.0737)	(0.0593)	(0.136)	(0.221)	(0.0445)	(0.0555)

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 4
Continued

Bank Financing of Working Capital (Continuous Variable)							0.129*	0.219***	0.0594	0.0458	0.233***	0.106**
							(0.0667)	(0.0546)	(0.130)	(0.204)	(0.0462)	(0.0537)
Observations	1,023	931	720	352	1,013	703	1,376	1,216	956	453	1,366	934
R-Squared	0.060	0.094	0.035	0.130	0.198	0.100	0.044	0.100	0.028	0.088	0.181	0.085

*Significant at 10%; **significant at 5%; ***significant at 1%.

The estimated model in Panel A is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha + \beta_1$Bank Dummy + β_2Small + β_3Medium + β_4Large + β_5Very Large + β_6Mid-Age + β_7Old + β_8Corporations + β_9Collectives + β_{10}State Ownership + β_{11}Competition Dummies + β_{12}City Dummies. The estimated model in Panel B is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha + \beta_1$Bank Financing of New Investments (or Working Capital) + β_2Dummy for 100% Bank Financing of New Investments (or Working Capital) + β_3Small + β_4Medium + β_5Large + β_6Very Large + β_7Mid-Age + β_8Old + β_9Corporations + β_{10}Collectives + β_{11}State Ownership + β_{12}Competition Dummies + β_{13}City Dummies. Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002 and from 1999 to 2002. Labor Productivity Growth is defined as the log change in productivity where productivity is (Sales−Total Material Costs)/Total Number of Workers. Labor Productivity Growth is computed from 2001 to 2002 and from 1999 to 2002. Growth in Total Factor Productivity (TFP) is computed from 2001–2002 where TFP is calculated as the residual from a regression of value added on labor and fixed assets, allowing for industry-specific coefficients for each variable. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Bank Dummy takes the value of 1 if the firm states that it has a current loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm said that it had no bank loan and had no overdraft facility or line of credit. Bank Financing of New Investments (Working Capital) is a continuous variable that measures the proportion of total financing of new investments (working capital) that is bank financed. The Dummy for 100% Bank Financing of New Investments (Working Capital) takes the value of 1 if the proportion of bank financing in total financing of new investments (working capital) is 100% and 0 otherwise. Firm Size Dummies are quintiles of total firm sales in 1999. Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. The omitted age dummy is less than five years. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperatives/Collectives Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4–6 Competitors, 7–15 Competitors, 16–100 Competitors, and Over 100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1–3 Competitors. Columns (1)–(6) in Panel A and columns (1)–(12) in Panel B present results for the full sample. Columns (7)–(12) in Panel A present results for a sample of firms that do not include firms registered as publicly traded firms and state-owned enterprises. We use OLS regressions with robust standard errors.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Columns (1)–(6) in table 4 show that the percentage of bank financing of new investments is associated with higher sales growth over 2001–2002 and 1999–2002 and productivity growth over 1999–2002. Productivity growth rates over the short term, and profit reinvestment rates in 2002, are positive but not significant. Interestingly, a 100% financing of new investments from banks is negative and significantly associated with profit reinvestment rates and sales growth over 1999–2002. This is to be expected, since firms with 100% bank financing include both firms that are doing exceptionally well as well as firms that are severely constrained or in distress. We find similar results with working capital in columns (7)–(12). Bank financing of working capital is positively associated with sales growth over 2001–2002 and 1999–2002, long-term productivity growth over 1999–2002, and profit reinvestment rates in 2002. In addition, financing 100% of working capital from banks is negatively associated with sales growth, profit reinvestment rates, and productivity growth over 1999–2002. These results hold for a subsample consisting of only private firms and also when microfirms are dropped from the sample. The results indicate that, in our sample, bank financing of both working capital and investment is associated with faster growth.

To check whether our results are driven by outliers, we perform several robustness checks. We have reestimated all our specifications by removing potential outliers (growth rates in excess of $\pm 1000\%$), by winsorizing the top 1% of the sales growth, reinvestment rate, and productivity growth variables, and by using median regressions. Our results (not reported) remain unchanged in all cases. The coefficients of *Bank Dummy* are qualitatively similar to those in table 4, indicating that our earlier results are not driven by outliers. It may be noted that our results on bank financing are not contaminated by the definition of financing from Other Sources. First, we use a dummy to capture whether the firm has a bank loan or not, which has no ambiguity in its definition. Second, as a robustness check, when we restrict the sample to certain cities with very low financing from Other Sources ($\leq 30\%$), we find our results unchanged. (We imposed a cut-off of 30% for the financing from Other Sources, since only one city [Wenzhou] has financing from Other Sources below 20%. So, our sample was restricted to five cities with $\leq 30\%$ financing from Other Sources: Wenzhou, Benxi, Hangzhou, Kunming, and Nanning.)

4.2 Bank finance and firm performance—treatment effects

In table 5, we investigate the relation between bank financing and sales growth, profit reinvestment, and productivity growth, controlling for the selection of firms that obtain bank financing. The estimates of the selection equation (even numbered columns) indicate that firms that obtain formal financing are more likely to have collateral that they can pledge and are large companies organized as corporations, with some but relatively few (<6) competitors. Interestingly, although firms in highly competitive industries grow more slowly, there is little

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Table 5
Bank financing and firm performance—selection model

	1	2	3	4	5	6	7	8	9	10	11	12
	Full sample						Drop public corporations and state-owned companies					
	Sales Growth [2001–2002]	Selection Equation	Profit Reinvestment Rate in 2002	Selection Equation	Labor Productivity Growth [2001–2002]	Selection Equation	Sales Growth [2001–2002]	Selection Equation	Profit Reinvestment Rate in 2002	Selection Equation	Labor Productivity Growth [2001–2002]	Selection Equation
Bank Dummy	0.310***		0.183***		0.116		0.244**		0.156**		0.068	
	[0.116]		[0.058]		[0.147]		[0.124]		[0.063]		[0.143]	
Collateral		0.968***		0.964***		0.941***		1.120***		1.102***		1.103***
		[0.076]		[0.080]		[0.089]		[0.095]		[0.099]		[0.107]
Small	−0.229***	0.101	0.075**	0.081	−0.101	−0.044	−0.219***	0.01	0.093***	−0.018	−0.129	−0.131
	[0.062]	[0.147]	[0.031]	[0.154]	[0.088]	[0.182]	[0.067]	[0.163]	[0.034]	[0.171]	[0.089]	[0.199]
Medium	−0.274***	0.240*	0.065**	0.224	−0.062	0.101	−0.276***	0.139	0.082**	0.142	−0.11	0.01
	[0.062]	[0.141]	[0.031]	[0.147]	[0.085]	[0.171]	[0.069]	[0.159]	[0.035]	[0.167]	[0.087]	[0.187]
Large	−0.289***	0.401***	0.03	0.363**	−0.147*	0.215	−0.284***	0.281*	0.06	0.262	−0.190**	0.085
	[0.065]	[0.140]	[0.032]	[0.146]	[0.086]	[0.168]	[0.073]	[0.161]	[0.037]	[0.167]	[0.088]	[0.186]
Very Large	−0.420***	0.850***	0.042	0.860***	−0.12	0.700***	−0.386***	0.742***	0.080*	0.714***	−0.116	0.610***
	[0.076]	[0.144]	[0.038]	[0.150]	[0.099]	[0.173]	[0.089]	[0.173]	[0.045]	[0.181]	[0.105]	[0.200]
Mid-Age	−0.001	−0.042	−0.007	−0.06	−0.097	−0.15	−0.022	−0.051	−0.014	−0.059	−0.1	−0.129
	[0.049]	[0.104]	[0.024]	[0.108]	[0.065]	[0.123]	[0.054]	[0.117]	[0.027]	[0.122]	[0.066]	[0.137]
Old	−0.033	−0.02	−0.061**	−0.075	−0.052	−0.177	−0.018	−0.184	−0.105***	−0.173	0.024	−0.314
	[0.061]	[0.129]	[0.030]	[0.135]	[0.080]	[0.154]	[0.077]	[0.169]	[0.039]	[0.174]	[0.092]	[0.194]
Corporations	−0.036	0.170*	0.070***	0.139	−0.06	0.147	−0.065	−0.026	0.111***	−0.023	−0.084	−0.064
	[0.044]	[0.089]	[0.022]	[0.094]	[0.055]	[0.102]	[0.050]	[0.108]	[0.025]	[0.113]	[0.060]	[0.123]
Cooperative	−0.150***	−0.02	0.024	0.003	−0.01	0.05	−0.207***	−0.063	0.061*	−0.046	−0.044	−0.052
	[0.056]	[0.124]	[0.028]	[0.130]	[0.072]	[0.144]	[0.066]	[0.148]	[0.034]	[0.154]	[0.079]	[0.166]
State	−0.032	0.062	−0.01	0.102	0.066	0.097						
	[0.049]	[0.102]	[0.025]	[0.108]	[0.067]	[0.126]						
4–6 Competitors	0.015	0.446***	0.032	0.462***	−0.079	0.412**	−0.021	0.554***	0.062	0.545***	−0.059	0.610***
	[0.080]	[0.161]	[0.040]	[0.170]	[0.099]	[0.182]	[0.095]	[0.200]	[0.048]	[0.207]	[0.109]	[0.220]
7–15 Competitors	−0.011	0.203	−0.004	0.236	−0.097	0.245	−0.034	0.276	0.014	0.317	−0.072	0.392*
	[0.074]	[0.155]	[0.037]	[0.163]	[0.092]	[0.175]	[0.090]	[0.198]	[0.045]	[0.204]	[0.104]	[0.217]

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 5
Continued

16–100 Competitors	−0.055	0.232	0.003	0.269*	−0.1	0.201	−0.059	0.291	0.041	0.249	−0.122	0.297
	[0.068]	[0.145]	[0.034]	[0.153]	[0.085]	[0.164]	[0.082]	[0.185]	[0.041]	[0.190]	[0.093]	[0.201]
> 100 Competitors	−0.146**	0.141	0.007	0.155	−0.153*	0.173	−0.134*	0.237	0.02	0.231	−0.143	0.308
	[0.064]	[0.140]	[0.032]	[0.147]	[0.083]	[0.161]	[0.078]	[0.177]	[0.039]	[0.183]	[0.091]	[0.197]
Hazard Lambda	−0.146**		−0.068*		−0.095		−0.122		−0.042		−0.059	
	[0.072]		[0.036]		[0.092]		[0.078]		[0.039]		[0.090]	
Observations	1,549	1,549	1,397	1,397	1,089	1,089	1,110	1,110	1,004	1,004	819	819

*Significant at 10%; **significant at 5%; ***significant at 1%.

A two-step selection model is used to estimate the effect of the endogenous Bank Dummy (binary treatment) on Firm Performance. The first step is: Bank Dummy = $\alpha_0 + \beta_1$Collateral + β_2Small + β_3Medium + β_4Large + β_5Very Large + β_6Mid-Age + β_7Old + β_8Corporations + β_9Collectives + β_{10}State Ownership + β_{11}Competition Dummies + β_{12}City Dummies. The second step is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha_1 + \gamma_1$Bank Dummy + γ_2Small + γ_3Medium + γ_4Large + γ_5Very Large + γ_6Mid-Age + γ_7Old + γ_8Corporations + γ_9Collectives + γ_{10}State Ownership + γ_{11}Competition Dummies + γ_{12}City Dummies.

Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002. Labor Productivity Growth is defined as the log change in productivity where productivity is (Sales − Total Material Costs)/Total Number of Workers. Labor Productivity Growth is computed from 2001 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Bank Dummy takes the value of 1 if the firm states that it has a current loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm said it had no bank loan and had no overdraft facility or line of credit. The identifying variable is Collateral, which is a dummy variable that takes the value of 1 if the financing required a collateral or a deposit and 0 if the financing did not require collateral or if the firm did not apply for a loan because of stringent collateral requirements or if the firm was rejected for a loan because of the lack of collateral. Firm Size Dummies are quintiles of total firm sales in 1999. Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. The omitted age dummy is less than five years. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperatives/Collectives Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4–6 Competitors, 7–15 Competitors, 16–100 Competitors, and Over 100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1–3 Competitors. Columns (1)–(6) present results for the full sample, and columns (7)–(12) present results for a sample of firms that do not include firms registered as publicly traded firms and state-owned enterprises. Hazard lambdas are reported for each of the second stages.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

evidence that banks lend to them at a lower rate. The hazard lambda (or the Inverse Mills Ratio), which measures the impact of self-selection, is reported at the foot of table 5.

In table 5, controlling for selection, the coefficients of *Bank Financing* are greater than in table 4, for both the full sample and the private firms subsample. This is in part because banks lend disproportionately to large firms, which grow more slowly than small firms. Once this selection effect is controlled for, there is a stronger relation between bank financing and firm growth. Similarly, the coefficients of *Bank Financing* in the productivity regressions are larger, albeit statistically insignificant. Thus, as in table 4, there is no evidence that bank financing is associated with inefficient growth.[40]

In unreported results, we perform several robustness checks (available in the Supplementary Data in tables A2 and A3 and figure 4). First, we use an expanded selection model where we consider a wider range of determinants of access to the formal financial sector. We find evidence that firms which report government help in obtaining loans, report receiving loans from other firms in their group, or which are located in export processing zones are more likely to receive loans. While affecting the probability of a loan, none of these variables significantly predict increases in the firm's growth rate, reinvestment rate, or productivity growth. Thus, there is no evidence that government help in obtaining loans is directed to firms that subsequently report better outcomes. However, we also find little evidence that firms which receive government help in obtaining loans perform less well than other firms. Controlling for these factors, we do not find evidence that the degree of perceived bank corruption, participation in loan guarantee programs, the educational level of the general manager, or reported political connections significantly affects the probability of obtaining a loan. Likewise, an index of property rights does not add explanatory power. However, the effect of the property rights could be subsumed by the size dummies, since larger firms in our sample report higher likelihood that the legal system will uphold their contract and property rights in business disputes.

We do find that firms which grew fast in 1999–2001 were more likely to have loans in 2001–2002. However, fast growth in 1999–2001 predicts slower growth in 2001–2002. Thus, while the banking system is more likely to lend to firms that grow fast, it is only those firms that the banks lend to that continue to grow fast.

Second, we also find that our results are robust to using a second class of treatment effects estimators, Propensity Score matching models (Rosenbaum and Rubin 1983), where the propensity score is the probability of a firm ob-

[40] The negative coefficients of the hazard rate lambdas in our second stage regressions suggest that firms which receive bank loans are predicted to perform more slowly than benchmark firms, perhaps as a result of unobservable private information. However, the coefficients are smaller than the positive coefficient of the bank loan dummy and for the most part not significant. Thus, the evidence suggests that bank financing is associated with enhanced growth.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

taining bank financing given a vector of firm characteristics (in our case *Size Dummies*, *Age Dummies*, *City Dummies*, *Corporations Dummy*, *Cooperatives Dummy*, and *State Ownership Dummy*) and is computed via a logit regression model. Conditional on observable characteristics, we find that bank-financed firms grow faster and reinvest more compared to nonbank-financed firms.

4.3 Bank finance and firm performance—robustness

4.3.1 Selection model robustness—access dummy. In Panel A of table 6, we repeat our analysis in table 5, with a broader definition of bank financing. *Access Dummy* is a dummy variable that takes the value of 1 if the firm had access to a bank loan in any year prior, from 1990 to 2001, and 0 otherwise. The results on *Sales Growth*, *Reinvestment Rate*, and *Productivity Growth* are qualitatively similar. One difference is that we find less evidence that bank access is skewed in favor of larger firms. This may be because small firms that do not have loans in 2001 may have obtained a loan in the preceding decade. Thus, these firms may have access to the formal sector but have a lower frequency of loans. All our results in Panel A hold when we look at a smaller sample without publicly traded corporations and state-owned enterprises. We do not report these results in the interest of space.

4.3.2 Government help. One of the remaining econometric concerns is that the association we observe between bank financing and firm performance is due to directed credit, that is, the Chinese government directs banks to lend to certain firms with good credit ratings, thus affecting both the selection decision and the outcome variable. It may be noted that when we control for firms that report government help in obtaining bank financing in both stages in the treatment effects model in table 6, we find that while the government help dummy predicts which firms get bank financing, government help in obtaining bank financing has no impact on firm performance.

We further deal with the issue of government help in two ways: first, we drop firms that report that they had government help in obtaining bank financing (15% of the overall sample) in column (1) of panel B of table 6 and find our results to be stronger. Bank-financed firms grow 10% faster than firms that do not have access to bank loans. This effect is larger than in the full sample results in table 4, where we found that bank-financed firms grew an average 7.5% faster than nonbank-financed firms. Hence, we have evidence that the positive association between bank finance and growth is not being driven by the firms that the government instructs the banks to lend to. We find similar effects when we look at the effect of bank financing on profit reinvestment rates and productivity growth but do not report these results in the interest of space.

Second, we use the *Government Help* variable explicitly in our identification strategy. Columns (2)–(4) of Panel B of table 6 presents the results of our in-

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Table 6
Bank financing and firm performance—robustness

Panel A: Selection model robustness—access to finance

	1	2	3	4	5	6
	Sales Growth [2001–2002]	Selection Equation	Profit Reinvestment Rate in 2002	Selection Equation	Labor Productivity Growth [2001–2002]	Selection Equation
Access Dummy	0.842**		0.652**		0.618	
	[0.403]		[0.262]		[0.455]	
Collateral		0.263***		0.221***		0.289***
		[0.072]		[0.076]		[0.085]
Hazard Lambda	−0.523**		−0.4**		−0.413	
	[0.244]		[0.158]		[0.276]	
Observations	1,566	1,566	1,413	1,413	1,102	1,102

Panel B: Government help

	1	2	3	4
	OLS Drop firms that report government help	Instrumental variables full sample		
	Sales Growth [2001–2002]	Sales Growth [2001–2002]	Sales Growth [2001–2002]	Sales Growth [2001–2002]
Instruments	–	Collateral	Government Help	Collateral and Government Help
Bank Dummy	0.100**	0.324***	0.080	0.296**
	[0.0398]	[2.675]	[0.398]	[2.527]
Observations	1799	1,549	2,117	1,531
R-squared	0.033			
Durbin–Wu–Hausman			4.2825	0.0003
Chi-Sq Test		(0.0385)	(0.9851)	(0.0609)
		157.63	50.67	87.55
First-Stage *F*-Stat		(0.000)	(0.0000)	(0.0000)
				0.662
Overidentification test:		–	–	(0.4157)
Conditional LR (CI, *p*-value)		[.0871, .5726] (0.0074)	[-.3283, .4895] (0.6930)	[.0656, .5358] (0.0119)

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 6
Continued

Panel C: Firms that did not apply for a bank loan

	1	2	3	4	5	6
	Full sample			Drop public corporations and state-owned companies		
	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]
Did Not Apply for a Bank Loan	−0.084**	−0.074***	0.017	−0.076*	−0.079***	−0.01
	(0.043)	(0.024)	(0.066)	(0.044)	(0.029)	(0.072)
Observations	1,432	1,271	9,45	1,048	922	736
R-squared	0.046	0.078	0.032	0.054	0.101	0.050

*Significant at 10%; **significant at 5%; ***significant at 1%.

In Panel A, a two-step selection effects model is used to estimate the effect of the endogenous Access Dummy (binary treatment) on Firm Performance. The first step is: Access Dummy = $\alpha_0 + \beta_1$Collateral + β_2Small + β_3Medium + β_4Large + β_5Very Large + β_6Mid-Age + β_7Old + β_8Corporations + β_9Collectives + β_{10}State Ownership + β_{11}Competition Dummies + β_{12}City Dummies. The second step is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha_1 + \gamma_1$Access Dummy+ γ_2Small + γ_3Medium + γ_4Large + γ_5Very Large + γ_6Mid-Age + γ_7Old + γ_8Corporations + γ_9Collectives + γ_{10}State Ownership + γ_{11}Competition Dummies + γ_{12}City Dummies.

In Panel B, in column (1), we report OLS regressions similar to column (1) of Table 4 but for a reduced sample of firms, which report no government help in obtaining bank finance. In columns (2)–(4), two-stage Instrumental Variable regressions are used. The first-stage regression is: Bank Dummy = $\alpha_0 + \beta_1$Collateral + β_2Government Help dummy + β_3Small + β_4Medium + β_5Large + β_6Very Large + β_7Mid-Age + β_8Old + β_9Corporations + β_{10}Collectives + β_{11}State Ownership + β_{12}Competition Dummies + β_{13}City Dummies + e. The second stage is: Sales Growth = $\alpha_1 + \gamma_1$Bank Dummy (predicted value from the first stage) + γ_2Small + γ_3Medium + γ_4Large + γ_5Very Large + γ_6Mid-Age + γ_7Old + γ_8Corporations + γ_9Collectives + γ_{10}State Ownership + γ_{11}Competition Dummies + γ_{12}City Dummies + u.

In Panel C, we report OLS regressions similar to Table 4 using the following specification: Sales Growth = $\alpha_1 + \gamma_1$Did Not Apply for a Bank Loan+ γ_2Small + γ_3Medium + γ_4Large + γ_5Very Large + γ_6Mid-Age + γ_7Old + γ_8Corporations + γ_9Collectives + γ_{10}State Ownership + γ_{11}Competition Dummies + γ_{12}City Dummies + u.

Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Labor Productivity Growth is defined as the log change in productivity where labor productivity is (Sales − Total Material Costs)/Total Number of Workers. Labor Productivity Growth is computed from 2001 to 2002. Access Dummy is a dummy variable that takes the value of 1 if the firm had access to a bank loan in any year prior from 1990 to 2001 and 0 otherwise. Bank Dummy takes the value of 1 if the firm said it had a loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm said it had no bank loan and had no overdraft facility or line of credit. The identifying variable in Panel A and columns (2) and (4) of Panel B is Collateral, which is a dummy variable that takes the value of 1 if the financing required a collateral or a deposit and 0 if the financing did not require collateral or if the firm did not apply for a loan because of stringent collateral requirements or if the firm was rejected for a loan because of the lack of collateral. In columns (1) and (4) of Panel B, we also use Government Help dummy as an identifying variable, which takes the value of 1 if a government agency or official assisted the firm in obtaining bank financing. Firm Size Dummies are quintiles of total firm sales in 1999. Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperatives/Collectives Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4–6 Competitors, 7–15 Competitors, 16–100 Competitors, and Over 100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1–3 Competitors. Did Not Apply for a Bank Loan takes the value of 1 if the firm reported not applying for a bank loan and 0 if the firm reported it had a bank loan.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

strumental variable (IV) estimation where we first instrument for *Bank Dummy* using *Collateral*, next using *Government Help*, and finally using both *Collateral* and *Government Help*. All the estimates reported are LIML estimates, which are more reliable than 2SLS estimates in the presence of weak instruments. When we instrument for *Bank Dummy* using *Collateral*, the results are as expected, and we find that bank financing is positively and significantly associated with firm growth. Note that the IV is similar to the treatment effects model in table 5 except that the latter is a control function approach where we incorporate the Inverse Mills Ratio. When we use *Government Help* as an instrument for *Bank Dummy*, we find that bank financing is no longer a significant predictor of firm growth. However, when we use both *Collateral* and *Government Help* as instruments, we find that *Bank Dummy* has a positive and significant effect on firm growth.

In all three cases, the instruments we use pass various tests of weak instruments. Both *Collateral* and *Government Help* are strongly positively correlated with *Bank Dummy* in the first stage (not reported), and the first-stage F-stats are larger than 10 (Staiger and Stock 1997 suggest that a first-stage F-stat of less than 10 for a single instrument may be used as a rule of thumb for identifying a weak instrument). The Durbin–Wu–Hausman endogeneity tests in columns (2) and (4) show that the OLS estimates are not consistent, and we need IV estimates. The Overidentification test in column (4) is not rejected, suggesting that the instruments are valid. In each column, we also report the conditional likelihood ratio confidence region and p-value proposed by Moreira (2003), both of which are robust to potentially weak instruments. The p-values reported test the null hypothesis that the bank dummy coefficient is zero using the conditional likelihood ratio test.

These results show conclusively that the positive association between bank financing and sales growth in our study is not being driven by Chinese government policy, which might instruct banks to provide financing to certain firms. In fact, we have evidence to the contrary suggesting that the firms which receive government help in obtaining bank financing do not grow as fast as firms which report no government help.

4.3.3 Firms that do not apply for bank loans. Firms in our survey report if they did not apply for a bank loan due to the following reasons: do not need loans, application procedures too cumbersome, stringent collateral requirements, high interest rates, bank corruption, and did not expect to be approved. In this section, we investigate if the firms that do not apply for bank loans due to any of the reasons listed above perform better than firms with a bank loan. In Panel C of table 6, we replicate table 4 using a dummy variable, *Did Not Apply for a Bank Loan*, which takes the value of 1 if the firm reported not applying for a bank loan and 0 if the firm has a bank loan. If the dummy variable was positive and significant, then it would mean either that

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

these firms do not have capital needs or that they are growing fast by sourcing their financing from nonbank sources.

Our results in Panel C of table 6 show that the firms that do not apply for a bank loan grow significantly slower and reinvest significantly less than firms with a bank loan. The *Did Not Apply for a Bank Loan* dummy is insignificant in the labor productivity regressions. In unreported results, we find that none of the reasons stated for not applying for a bank loan are ever positive and significant in any of the regression specifications. The firms in our sample also report if their application for a bank loan was rejected. Here again, we find no evidence that firms rejected for a bank loan performed better than firms that had access to a bank loan.

4.4 Financing of new investments and working capital—banks versus informal finance

In table 7 of Panel A, we examine the association between indicators of performance and bank financing for those firms that explicitly report using bank financing to fund at least some (>0%) of their new investments or working capital. The coefficient estimates for the *Bank Financing Dummy* are shown in Panel A. Since we have data on uses of bank financing only for 2002, we do not estimate a selection model. We interpret our estimates as a consistency check on our earlier results rather than as evidence of a causal relationship.

Panel A of table 7 shows that our earlier results hold for firms that report heavy use of bank financing to fund investment and working capital. Such firms tend to grow faster and reinvest more of their profits than comparable firms. This faster growth is not associated with slower improvements in productivity.

In Panel B of table 7, we show the association between performance indicators and self-financing of investment and working capital. To recall, *Self-Financing1* takes the value of 1 if the firm reports that it has financed at least half of its new investments or working capital from Informal and Other Sources. *Self-Financing1* takes the value of 0 if the firm reports not using financing from Informal or Other Sources to fund investment or working capital. To be consistent with Allen, Qian, and Qian (2005), we combine internal sources with external informal sources in defining *Self-Financing2*. *Self-Financing2* takes the value of 1 if the firm reports using Internal, Informal, Loans from Family and Friends, and Other financing to fund at least half of its new investments or working capital. *Self-Financing2* takes the value of 0 if the firm reports not using these sources to fund investment or working capital. Thus, *Self-Financing1*, by excluding internal financing, places more weight on "external" informal financing, whereas *Self-Financing2* also includes funds explicitly designated as retained earnings or contributed by family members.[41]

[41] In a sensitivity analysis on the *Self-Financing1* variable, we find that all our results are robust to not including the hurdle rate of 50% for working capital in the definition of *Self-Financing1*, as well as to varying the hurdle rate of informal financing of working capital over 50%.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Table 7
Financing proportions of new investments and working capital—bank vs. informal financing

Panel A: Bank financing

	1	2	3	4	5	6
	Full sample			Drop public corporations and state-owned companies		
	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]
Bank Financing Dummy	0.092*	0.07**	0.045	0.052	0.074*	0.06
	[0.056]	[0.031]	[0.076]	[0.058]	[0.038]	[0.077]
Observations	895	809	621	659	592	480
R-squared	0.063	0.099	0.039	0.068	0.103	0.058

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Table 7
Continued

Panel B: Self-financing

	1	2	3	4	5	6	7	8	9	10	11	12
	Full sample						Drop public corporations and state-owned companies					
	Sales Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Labor Productivity Growth [2001–2002]	Sales Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Labor Productivity Growth [2001–2002]
Self-Financing1 (Informal + Other)	−0.04		−0.126***		0.092		−0.026		−0.127***		0.084	
	[0.039]		[0.020]		[0.059]		[0.050]		[0.024]		[0.064]	
Self-Financing2 (Informal + Other + Internal + Loans from Family and Friends)		−0.015		0.045*		0.154**		0.006		0.043		0.151**
		[0.048]		[0.024]		[0.072]		[0.058]		[0.030]		[0.075]
Observations	1,363	1,372	1,203	1,207	952	956	984	992	874	877	719	722
R-squared	0.056	0.055	0.110	0.103	0.041	0.042	0.06	0.058	0.111	0.104	0.048	0.05

*Significant at 10%; **significant at 5%; ***significant at 1%.

The estimated model is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha + \beta_1$Bank Financing or Self-Financing1 or Self-Financing2 + β_2Small + β_3Medium + β_4Large + β_5Very Large + β_6Mid-Age + β_7Old + β_8Corporations + β_9Collectives + β_{10}State Ownership + β_{11}Competition Dummies + β_{12}City Dummies.

Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002. Labor Productivity Growth is defined as the log change in productivity where labor productivity is (Sales − Total Material Costs)/Total Number of Workers. Labor Productivity Growth is computed from 2001 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Bank Financing Dummy takes the value of 1 if the firm states that it has a loan from a bank or financial institution and an overdraft facility or line of credit and reports that bank finances at least 50% of new investments or working capital. Bank Financing takes the value of 0 if the firm states that it has no loan from a bank or financial institution and said it had no overdraft facility or line of credit and bank financing of both new investments and working capital was equal to 0%. Self-Financing1 takes the value of 1 if the sum of financing of either new investments or working capital from Informal and Other Sources is greater than 50%. Self-Financing1 takes the value of 0 if the sum of financing of new investments and working capital from Informal and Other Sources is equal to 0%. Self-Financing2 takes the value of 1 if the sum of financing of new investments or working capital from Retained Earnings, Informal, Loans from Family and Friends, and Other Sources is greater than 50%. Self-Financing2 takes the value of 0 if the sum of financing of new investments or working capital from Retained Earnings, Informal, Loans from Family and Friends, and Other Sources is equal to 0%. Firm Size Dummies are quintiles of total firm sales in 1999. Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperatives/Collectives Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4–6 Competitors, 7–15 Competitors, 16–100 Competitors, and Over 100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1–3 Competitors. We use OLS regressions with robust standard errors.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 202

Panel B of table 7 shows that there is no association between firm growth and either measure of self-financing. The coefficients of both measures of self-financing are negative although not significant in the sales growth regression. Interestingly, firms that are self-funded primarily through informal or other sources reinvest a lower proportion of their earnings in their business. However, the negative association between self-financing and profit reinvestment rates switches to positive when we incorporate internal and family funds as part of self-financing sources. We find that it is the inclusion of internal financing (rather than family funds) that drives the positive association between self-financing and profit reinvestment rates.

In unreported results, when we look at the sample of firms that did not apply for a bank loan or who were rejected for a loan by the bank, we find no evidence that firms relying mostly on informal or other financing grow faster or reinvest more than firms using other sources of financing. The coefficients of *Self-Financing1* and *Self-Financing2* are negative (but not significant) in the sales growth regressions and negative and significant at the 1% level in the profit reinvestment rate regressions. If we include the individual components of self-financing in the regressions, we find that firms that use more financing from Informal and Other Sources reinvest significantly less than the firms that rely on their internal funds. There is also evidence suggesting that firms relying on financing from Informal and Other Sources grow slower, though not significantly, than firms relying on their Retained Earnings.[42]

Overall, these results suggest that any positive association between nonbank financing and profit reinvestment rates stems from the use of retained earnings rather than informal sources and other alternative financing channels. Our results on internal financing are consistent with recent work (e.g., Guariglia, Liu, and Song 2008) showing that the growth of Chinese firms is fostered by the availability of internal financing.

When we use the broader definition of informal financing, *Self-Financing2*, we also find that self-funded firms report higher growth in productivity. In unreported results, we interact the self-financing variable with size quintile dummies and find that it is the smallest quintile of firms (which we know have limited access to bank finance) that report productivity increases when their primary financing sources include internal, informal, family, or other financing. Thus, informal financing might be the optimal financing source for such firms, paralleling the use of angel financing in the United States.

5. Discussion

In this section, we discuss our findings in the context of recent evidence on the link between financing and growth in China. Allen, Qian, and Qian (2005) find

[42] The association between slow growth and dependence on informal and other sources is significant and strong when we look at growth over the longer time period of 1999–2002.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

that while the private sector dominates the state and listed sectors in both the size of the output and the growth trend, there is a huge reliance on nonbank financing sources among private sector firms. Given the weak external markets and poor legal protection of minority and outside investors in China, Allen, Qian, and Qian (2005) attribute the growth of the private sector to the reliance on alternative financing channels and corporate governance mechanisms, such as reputation and relationships. They interpret this as evidence against the finance and growth literature that the development of stock markets and a banking system are important for growth of firms and economies. However, due to data limitations, the results in Allen, Qian, and Qian (2005) are based on an analysis of the seventeen largest, and perhaps unrepresentative, firms in the two most developed regions in the country. Further, their definition of informal financing channels includes retained earnings, informal financing, equity issuance, and all other sources of funds except bank loans, state funding, and foreign investment.

While our data on twenty-four hundred firms (including 1,720 nonpublicly traded and nonstate companies) confirm the wide use of financing channels other than bank financing, our regression results suggest that it is the formal financing channel, specifically bank financing, that is positively associated with higher growth and reinvestment. These increases are not associated with decreases in firm productivity. We find no evidence that alternative financing channels such as informal sources have a positive impact on growth and reinvestment or that they substitute for the formal sector. The arguments in Allen, Qian, and Qian (2005) can best be interpreted as a second best: how finance can cope with restrictions on entry and with pervasive state ownership of banks.

To investigate the relation between growth and bank financing further, we estimate a selection model that takes into account that banks may use proprietary information not observed by researchers to allocate credit to firms that subsequently grow faster. Controlling for this selection bias strengthens the relation between bank financing and subsequent growth.

Our results are consistent with other studies emphasizing the role of institutions and formal financing in China. Cull and Xu (2005) find that profit reinvestment rates are affected by enterprise managers' perceptions about the security of property rights, the risk of expropriation by government officials, the efficiency and reliability of courts, and access to credit. In a more recent paper, Cull, Xu, and Zhu (2009) find that despite a biased and inefficient banking system, trade credit does not play an economically significant role in China. There are also more recent studies emphasizing the link between institutions and growth at the provincial level. Cheng and Degryse (2006) explore the impact of the development of bank versus nonbank financial institutions on the growth rate of Chinese provinces over the period 1995–2003 and conclude that only bank loans have a significant impact on local economic growth. Fan et al. (2009) find that inward FDI within China flows disproportionately into

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

provinces with less corrupt governments and governments that better protect private property rights.[43]

So how do we reconcile our results with the inefficiencies in the formal financial system described in Section 1? The formal financial system does serve a small sector of the economy, and both the aggregate statistics and firm responses suggest that there are imperfections in the allocation of capital. However, our results show that despite these weaknesses, at the margin, private sector firms that have loans from the formal sector do better than other firms, and having a better-developed financial system will only bring more benefits. Overall, our claim is not that the current Chinese growth experience is driven by banks (we do not have the data to show this) but that formal financing through banks is associated with higher growth of Chinese firms rather than financing from informal channels.

We have no evidence that the firms depending on external financing of their investment through informal channels are the fastest-growing firms. Rather, the evidence suggests that, if anything, retained earnings are associated with better performance than external informal financing and that firms that have bank loans grow faster than firms relying on informal or other sources. However, even though in the aggregate informal financing sources are not associated with high growth, the low reliance of very small firms on bank financing suggests that self-financing is important for the growth and survival of small entrepreneurial firms. For some of these firms, informal financing may play a role akin to the critical role played by angel finance in the financing and creation of rapid-growth start-ups in developed economies. Thus, while the informal sector may have its own niche in financing very small firms, we find no evidence that it scales up and is an efficient substitute for bank financing for most firms.

6. Conclusion

With one of the largest and fastest-growing economies in the world, China provides many puzzles. More recently, it has been singled out as a counterexample to the findings of the finance and growth literature: While Chinese financial systems and institutions are underdeveloped, its economy, particularly its private sector, has been growing very fast. One explanation for this observation has been that China has alternative mechanisms and institutions that play an important role in supporting its growth and are good substitutes

[43] Fan et al. (2009) estimate a cross-country (without China) FDI model explaining inward FDI using different measures of institutions and make out-of-sample predictions from their model for China. They find that China is no exception, since Chinese FDI inflows are in line with what the model predicts for a country at China's level of institutional development as measured by general government quality and rule of law. However, when they measure institutions by "strength of executive constraints," they find that China receives more FDI than predicted by their model. But, they argue that this could be because of other reasons, including underestimation of the strength of checks on executive power in China or foreign firms enjoying better protections than identical Chinese firms.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

for standard corporate governance mechanisms and financing channels (Allen, Qian, and Qian 2005).

In this article, we take a closer look at the financing of twenty-four hundred Chinese firms and their performance, using a detailed firm-level survey. We find that in China, private firms' use of formal financing channels is comparable to its use in other developing countries, and it is financing from these sources that is positively associated with firm performance. Contrary to earlier findings, fund-raising from informal channels is not associated with faster firm growth. To the extent that there are measurable benefits of informal financing, they arise only when retained earnings is classified as informal financing, as in some earlier studies. These findings have broader implications for the role of informal versus formal financial systems. While informal financing may provide advantages in financing very small entrepreneurial firms, as in developed economies, our results contradict the belief that nonstandard financing mechanisms provide effective substitutes to formal financing channels in promoting growth. Our findings question whether reputation and relationship-based financing are responsible for the growth of the fastest-growing firms in China.

Supplementary Data

Supplementary data are available online at http://rfs.oxfordjournals.org.

References

Allen, F., J. Qian, and M. Qian. 2005. Law, Finance, and Economic Growth in China. *Journal of Financial Economics* 77:57–116.

———. 2008. China's Financial System: Past, Present and Future. In L. Brandt and T. G. Rawski (eds.), *China's Great Economic Transition*, pp. 506–68. Cambridge, UK: Cambridge University Press.

Arnott, R., and J. E. Stiglitz. 1991. Moral Hazard and Nonmarket Institutions; Dysfunctional Crowding Out or Peer Monitoring? *American Economic Review* 81:179 –90.

Ayyagari, M., A. Demirguc-Kunt, and V. Maksimovic. 2009. Are Innovating Firms Victims or Perpetrators? Tax Evasion, Bribe Payments, and the Role of External Finance in Developing Countries. *SSRN Working Paper Series*.

Beck, T., A. Demirgüç-Kunt, and V. Maksimovic. 2005. Financial and Legal Constraints to Firm Growth: Does Size Matter? *Journal of Finance* 60:137–77.

———. 2006. The Influence of Financial and Legal Institutions on Firm Size. *Journal of Banking and Finance* 30:2995–3015.

Beck, T., A. Demirgüç-Kunt, L. Laeven, and R. Levine. 2008. Finance, Firm Size, and Growth. *Journal of Money, Credit, and Banking* 40:1379–405.

Bekaert, G., C. R. Harvey, and C. Lundblad. 2001. Emerging Equity Markets and Economic Development. *Journal of Development Economics* 66:465–504.

———. 2005. Does Financial Liberalization Spur Growth? *Journal of Financial Economics* 77:3–56.

Bell, C., T. N. Srinivasan, and C. Udry. 1997. Rationing, Spillover, and Interlinking in Credit Markets: The Case of Rural Punjab. *Oxford Economic Papers* 49:557–85.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Berger, A., and G. Udell. 1995. Relationship Lending and Lines of Credit in Small Firm Finance. *Journal of Business* 68:351–81.

———. 1998. The Economics of Small Business Finance: The Roles of Private Equity and Debt Markets in the Financial Growth Cycle. *Journal of Banking and Finance* 22:613–73.

———. 2006. A More Complete Conceptual Framework for SME Finance. *Journal of Banking and Finance* 30:2945–66.

Bertrand, M., A. Schoar, and D. Thesmar. 2007. Banking Deregulation and Industry Structure: Evidence from the French Banking Reforms of 1985. *Journal of Finance* 62:597–628.

Besley, T., and A. R. Levenson. 1996. The Role of Informal Finance in Household Capital Accumulation: Evidence from Taiwan (China). *Economic Journal* 106:39–59.

Bias, B., and C. Gollier. 1997. Trade Credit and Credit Rationing. *Review of Financial Studies* 10:903–37.

Bolton, P., and D. S. Scharfstein. 1996. Optimal Debt Structure and the Number of Creditors. *Journal of Political Economy* 104:1–25.

Bose, P. 1998. Formal–Informal Sector Interaction in Rural Credit Markets. *Journal of Development Economics* 56:265–80.

Boyreau-Debray, G. 2003. Financial Intermediation and Growth: Chinese Style. World Bank Policy Research Working Paper 3027.

Burkart, M., T. Ellingsen, and M. Giannetti. Forthcoming. What You Sell Is What You Lend? Explaining Trade Credit Contracts. *Review of Financial Studies*. Advance Access published November 24, 2008, 10.1093/rfs/hhn096.

Chen, C., and H. Shih. 2004. *Banking and Insurance in the New China*. Cheltenham, UK: Edward Elgar Publishing Limited.

Cheng, X., and H. Degryse. 2006. The Impact of Bank and Non-Bank Financial Institutions on Local Economic Growth in China. Tilburg Law and Economics Center Discussion Paper.

Cole, S. A. 2009. Financial Development, Bank Ownership, and Growth: Or, Does Quantity Imply Quality? *Review of Economics and Statistics* 91:33–51.

Cousin, V. 2006. *Banking in China*. New York: Palgrave Macmillan.

Cull, R., and L. C. Xu. 2000. Bureaucrats, State Banks, and the Efficiency of Credit Allocation: The Experience of Chinese State-Owned Enterprises. *Journal of Comparative Economics* 28:1–31.

———. 2003. Who Gets Credit? The Behavior of Bureaucrats and State Banks in Allocating Credit to Chinese State-Owned Enterprises. *Journal of Development Economics* 71:533–59.

———. 2005. Institutions, Ownership, and Finance: The Determinants of Profit Reinvestment among Chinese Firms. *Journal of Financial Economics* 77:117–46.

Cull, R., L. C. Xu, and T. Zhu. 2009. Formal Finance and Trade Credit during China's Transition. *Journal of Financial Intermediation* 18:173–92.

Delios, A., and Z. Wu. 2005. Legal Person Ownership, Diversification Strategy and Firm Profitability in China. *Journal of Management and Governance* 9:151–69.

Demirgüç-Kunt, A., and V. Maksimovic. 1998. Law, Finance, and Firm Growth. *Journal of Finance* 53:2107–37.

———. 1999. Institutions, Financial Markets, and Firms Debt Maturity. *Journal of Financial Economics* 54:295–336.

Diamond, D. 1984. Financial Intermediation and Delegated Monitoring. *Review of Economics Studies* L1:393–414.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Dollar, D., M. Hallward-Driemeier, A. Shi, S. Wallsten, S. Wang, and L. C. Xu. 2003. Investment Climate in China: A Tale of Five Cities. *Transition Letters* 14.

Dollar, D., S. Wang, L. C. Xu, and A. Shi. 2004. Improving City Competitiveness through the Investment Climate: Ranking 23 Chinese Cities. World Bank Working Paper Series.

Durnev, A., K. Li, R. Morck, and B. Yeung. 2004. Capital Markets and Capital Allocation: Implications for Economies in Transition. *Economics of Transition* 12:593–634.

Fan, J. P. H., J. Huang, and N. Zhu. 2008. Distress without Bankruptcy: An Emerging Market Perspective. SSRN Working Paper Series.

Fan, J. P. H., R. Morck, C. Xu, and B. Yeung. 2009. Institutions and Foreign Investment: China versus the World. *World Development* 37:852–65.

Farrell, D., S. Lund, J. Rosenfeld, F. Morin, N.Gupta, and E. Greenberg. 2006. Putting China's Capital to Work: The Value of Financial System Reform. McKinsey Global Institute Report.

Fenn, G., N. Liang, and S. Prowse. 1997. The Private Equity Market: An Overview. *Financial Markets, Institutions, and Instruments* 6:1–106.

Floro, M. S., and D. Ray. 1997. Vertical Links between Formal and Informal Financial Institutions. *Review of Development Economics* 1:34–56.

Garmaise, M. J., and T. J. Moskowitz. 2003. Informal Financial Networks: Theory and Evidence. *Review of Financial Studies* 16:1007–40.

Gomes, A. 2000. Going Public without Governance: Managerial Reputation Effects. *Journal of Finance* 55:615–46.

Gregory, N., and S. Tenev. 2001. The Financing of Private Enterprise in China. *Finance and Development* 38:14–17.

Guariglia, A., X. Liu, and L. Song. 2008. Is the Growth of Chinese Firms Constrained by Internal Finance? IZA Discussion Paper No. 3808.

Guiso, L., P. Sapienza, and L. Zingales. 2004a. The Role of Social Capital in Financial Development. *American Economic Review* 94:526–56.

———. 2004b. Does Local Financial Development Matter? *Quarterly Journal of Economics* 119:929–69.

Han, S. L. 2006. Secured Transactions Law Reform in China: Can a Commercial Law Serve the Needs of the Market. *China Law and Governance Review* 3, 1–8.

Hart, O., and J. Moore. 1995. Debt and Seniority: An Analysis of Hard Claims in Constraining Management. *American Economic Review* 85:567–87.

Heckman, J. J. 1979. Sample Selection Bias as a Specification Error. *Econometrica* 47:153–61.

Hellman, T., L. Lindsey, and M. Puri. 2008. Building Relationships Early: Banks in Venture Capital. *Review of Financial Studies* 21:513–41.

Hoff, K., and J. E. Stiglitz. 1998. Moneylenders and Bankers: Price-Increasing Subsidies in a Monopolistically Competitive Market. *Journal of Development Economics* 55:485–518.

Jain, S. 1999. Symbiosis vs. Crowding-Out: The Interaction of Formal and Informal Credit Markets in Developing Countries. *Journal of Development Economics* 59:419–44.

Johnson, S. H., J. McMillan, and C. M. Woodruff. 1999. Property Rights, Finance, and Entrepreneurship. *ISBN/Report Conference Paper*, Symposium in Economics.

Kandori, M. 1992. Social Norms and Community Enforcement. *Review of Economic Studies* 59:63–80.

King, R. G., and R. Levine. 1993. Finance and Growth: Schumpeter Might Be Right. *Quarterly Journal of Economics* 108:717–38.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Klapper, L. 2006. The Role of Factoring for Financing Small and Medium Enterprises. *Journal of Banking and Finance* 30:3111–30.

La Porta, R., F. Lopez-de-Silanes, and A. Shleifer. 2002. Government Ownership of Banks. *Journal of Finance* 57:265–301.

Lerner, J. 1998. "Angel" Financing and Public Policy: An Overview. *Journal of Banking and Finance* 22:773–83.

Levine, R., N. Loayza, and T. Beck. 2000. Financial Intermediation and Growth: Causality and Causes. *Journal of Monetary Economics* 46:31–77.

Levine, R., and S. Zervos. 1998. Stock Market, Banks, and Economic Growth. *American Economic Review* 88:537–58.

Li, J. 2005a. Zhejiang Minjian Jiedai Diaocha (Zhejiang Informal Finance Survey). *Huan Qui Cai Jing (Global Business and Finance)* 2:82–85.

———. 2005b. Zhongguo dixia jinrong guimo yu hongguan jingji yingxiang yanjiu (Research on the Scale of Underground Financing and Its Economic Effects). Beijing, China: Zhongguo jinrong chubanshe.

Li, K., and N. Prabhala. 2007. Self-Selection Models in Corporate Finance. In B. Espen Eckbo (ed.), *Handbook of Empirical Corporate Finance*, chapter 2, pp. 37–83. Amsterdam, The Netherlands: Elsevier.

Liu, T., and K. Li. 2001. Impact of Liberalization of Financial Resources in China's Economic Growth: Evidences from Provinces. *Journal of Asian Economics* 12:245–62.

Linton, K. 2006. Access to Capital in China: Competitive Conditions for Foreign and Domestic Firms. SSRN Working Paper Series.

Love, I. 2003. Financial Development and Financing Constraint: International Evidence from the Structural Investment Model. *Review of Financial Studies* 16:765–91.

Ma, Z. 2003. Rural Informal Finance. In C. Findlay, A. Watson, and E. Cheng (eds.), *Rural Financial Markets in China*, pp. 105–19. Canberra, Australia: Asia Pacific Press.

Moreira, M. J. 2003. A Conditional Likelihood Ratio Test for Structural Models. *Econometrica* 71:1027–48.

OECD. 2005. Economic Surveys–China. Paris, France: OECD.

Petersen, M., and R. G. Rajan. 1994. The Benefits of Firm–Creditor Relationships: Evidence from Small Business Data. *Journal of Finance* 49:3–37.

Rajan, R. G., and L. Zingales. 1998. Financial Dependence and Growth. *American Economic Review* 88:559–86.

Rosenbaum, P. R., and D. B. Rubin. 1983. The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika* 70:41–55.

Staiger, D., and J. H. Stock. 1997. Instrumental Variables Regression with Weak Instruments. *Econometrica* 65:557–86.

Stigltiz, J. E. 1990. Peer Monitoring and Credit Markets. *World Bank Economic Review* 4:351–66.

Straub, S. 2005. Informal Sector: The Credit Market Channel. *Journal of Development Economics* 78:299–321.

Tsai, K. 2002. *Back-Alley Banking: Private Entrepreneurs in China*. Ithaca, NY: Cornell University Press.

———. Imperfect Substitutes: The Local Political Economy of Informal Finance and Microfinance in Rural China and India. *World Development* 32:1487–507.

Uchida, H., G. F. Udell, and N. Yamori. 2006. SME Financing and the Choice of Lending Technology. RIETI Discussion Paper Series 06-E-025.

Udry, C. 1994. Risk and Insurance in a Rural Credit Market: An Empirical Investigation in Northern Nigeria. *Review of Economic Studies* 61:495–526.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022

Varghese, A. 2005. Bank–Moneylender Linkage as an Alternative to Bank Competition in Rural Credit Markets. *Oxford Economic Papers* 57:315–35.

Wang, X., L. C. Xu, and T. Zhu. 2004. State-Owned Enterprises Going Public: The Case of China. *Economics of Transition* 12:467–87.

World Bank–People's Bank of China (WB-PBOC). 2006. *Secured Transactions Reform and Credit Market Development in China.* Beijing, China: China CITIC Press.

Wurgler, J. 2000. Financial Market and the Allocation of Capital. *Journal of Financial Economics* 58:187–214.

Xu, X., and Y. Wang. 1997. Ownership Structure, Corporate Governance and Firm Performance: The Case of Chinese Stock Companies. World Bank Policy Research Paper #1794.

Zhang, X., and C. D. Booth. 2001. Chinese Bankruptcy Law in an Emerging Market Economy: The Shenzhen Experience. *Journal of Asian Law* 15:1–33.

Zhang, J., Z. Yuan, and P. Lin, 2002. From Interpersonal Credit to Unregulated, Private-Governed Financial Institutions: Formats and Pricing Mechanisms of Informal finance. *Journal of Financial Research* 268:101–9.

Zhou, L., and Z. Wang. 2002. Empirical Research of Regional Financial Development and Economic Growth in China (1978–2000). *Financial Research* 10:1–13.

Downloaded from https://academic.oup.com/rfs/article/23/8/3048/1588349 by Sichuan University user on 20 November 2022